AICTE PS 26261 — Corrected Specification v2.1, Competitive Research & Uniqueness Playbook

Project: AICTE Pre-Scrutiny, Document Authentication & Regulatory Compliance Platform (SIH 2026, PS 26261)
Applies to: your spec `v2.0.0` → this document defines `v2.1.0`
Prepared: 25 October 2026 (idea-submission deadline reported as 28 October 2026 on an unofficial mirror — confirm on sih.gov.in)
Companion to: the earlier verdict. This file is the full evidence base, the correction list, the new differentiators, and the plan.

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0. How to read this document

0.1 Evidence tags (use the same tags in your own deck and README)

Tag	Meaning
[V]	Verified from a source I opened (primary or a faithful copy of the primary).
[R]	Reported by a secondary source (news, college page, summary). Treat as probably right; confirm in the APH PDF before putting it in the deck.
[?]	Not verified. Appears in your v2.0.0 spec or in my memory; **must be checked against the APH 2024-27 PDF and the 2025-26 / 2026-27 addenda** before you cite it.
[A]	Assumption or design choice — ours, not AICTE's. Must be labelled as such.

0.2 What I did and did not do

Did: read the official PS 26261 brief (via a mirror of sih.gov.in), five public repositories/demos built for the same PS, secondary copies of AICTE faculty norms, AICTE-issued EoA letters (they carry useful clause references), news on the 2025-26 and 2026-27 addenda, and the DigiLocker requester API documentation.

Did not: open the APH 2024-27 PDF itself or either addendum PDF; read competitor source code beyond README level; survey commercial KYC/document-verification vendors in depth; test any competitor's software. Every competitor statement below is "per README / public demo".

Consequence: the regulatory numbers in Section 3 are split into **[V]/[R]** (safe to use) and **[?]** (verify first). Do not let a judge find the first wrong page number — your spec's selling point is *verifiable* citations.

0.3 Contents

1. Executive summary
2. Research: brief decomposition, context, existing solutions, drawbacks, capability matrix
3. Corrections to spec v2.0.0 (C-01 … C-32)
4. New unique additions (U-01 … U-12) + parity items (P-01 … P-08)
5. Specification v2.1 deltas: architecture, rule-pack layout, schema, API, UI
6. Corrected evaluation protocol (how to show *measured* results honestly)
7. Showing uniqueness: positioning, demo script, deck plan, jury Q&A, claims register
8. Build plan, priorities, cut list, risks
9. Appendices: sources, APH verification checklist, license table, glossary

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1. Executive summary

1.1 Verdict in six lines

1. **Concept:** not unique. PS 26261 itself defines the six statuses, the safeguards and the evaluation metrics; at least five public implementations already exist for this PS.
2. **Closest competitor:** *Pramana* (472 commits, 162 tests, real CCA-root signature checks, C2PA, revision "time machine", fine-tuned local extractor, encryption/retention). Do **not** try to out-forensic it in the time you have.
3. **Your real edge:** governance of the *regulation itself* (versioned rules, addenda, impact analysis), honest evaluation, and **remediation** (what would make this pass), not just detection.
4. **Biggest new finding:** the APH 2024-27 is a three-year handbook that is **changed by annual addenda** — e.g. the 2025-26 addendum reportedly removed "adjunct faculty" and set Professor-of-Practice to 20% (5% reserved for women) **[R]**, and the 2026-27 addendum reportedly raised supernumerary seats to 25% and requires "nil deficiency based on self-disclosure" **[R]**. Any engine with a single frozen rule file is wrong by design. Yours can be right by design.

5. **Your spec has accuracy and security problems** (cadre formula, adjunct caps now outdated, page-number conflicts, `/365.25` date bug, role chosen by the client at login, raw PAN, AGPL dependencies, no encryption/retention). All are fixable and are listed in Section 3.
6. **Benchmark numbers in v2.0.0 are targets, not results.** Present them as targets until measured, with confidence intervals and a false-accusation rate.

1.2 The five repositioning moves

#	Move	Why it works
1	**"Regulation-as-code with layered addenda"** — base handbook + dated overlays, effective-cycle resolution, per-value provenance	The handbook really changes mid-cycle; nobody else documents this.
2	**Impact simulator + maker-checker publish gate**	Answers "what happens to 50 pending applications if AICTE amends a norm?" before it goes live.
3	**Counterfactual remediation engine + Intake Planner**	Turns every deficiency into a quantified fix (hire N, reduce intake by M, upgrade to X Mbps). Detection-only tools stop at the flag.
4	**Disclosure Auditor (public-website three-way check)**	Uses a legal, public, AICTE-mandated disclosure source; no restricted DB needed.
5	**Honest evaluation: False-Accusation Rate, abstention, CIs, template shift, red-team matrix**	Adverse regulatory action harms institutions; a platform that measures its own harm is credible.

1.3 Scoreboard (self-assessment, 1–5)

Dimension	v2.0.0	v2.1 target	Note
Fit to PS brief	4	5	Add floor-plan evidence, application-history records, split 8 metrics
Regulatory accuracy	3	4	Fix cadre math, adjunct/PoP, verify page refs
Technical novelty	2	3	Novelty is in system design, not in detectors
Distinctiveness vs public repos	2	4	Moves 1–5 above
Security/privacy posture	2	4	Server-side roles, encryption, retention, sandbox
Evaluation credibility	2	4	Measured, CI'd, FAR, red-team
Build feasibility (team of ~4–6)	2	3	Cut list in Section 8

1.4 If you only have 3 days (until idea submission)

1. Rewrite the pitch around Moves 1–5 (Section 7.3 gives the slide plan).
2. Remove or relabel every unverified number (Section 9.2 checklist).
3. Fix the six embarrassing items: cadre formula, adjunct/PoP, `/365.25`, role-at-login, raw PAN, benchmark-as-target.
4. Add the two cheapest high-impact features to the *idea*: layered rule packs and remediation engine.
5. Add an honest **Verification Feasibility Matrix** (U-06) so you never claim access you don't have.

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2. Research

2.1 The official brief, decomposed into testable requirements

Source: official PS 26261 text **[V]** (S1). Left column = what the brief asks; right = status of v2.0.0 and the v2.1 action.

ID	Requirement in the brief	v2.0.0	v2.1 action
R01	Identify application category; apply the right checklist	Covered	Keep; add classifier confidence
R02	Use institutional records already in authorized AICTE systems: affidavits/undertakings, accreditation, NOCs, **previous EoA and approval status**, approved programmes/intake, faculty, compliance/deficiency status, **TER payment status, application history**	Partial	Add `eoa_history`, `application_history`, `accreditation_records` tables (C-27)
R03	OCR, document classification, extraction	Covered (keyword	Add text+layout classifier with confidence (P-05)

ID	Requirement in the brief	v2.0.0	v2.1 action
		classifier is weak)	
R04	Identify missing, incomplete, expired, inconsistent info	Covered	Separate **completeness** from the six verification statuses (C-09)
R05	Structured validation report; allow rectification where process permits	Covered	Keep; add rectification-scope check (flag edits outside requested scope)
R06	Flags with evidence and reason; "preliminary indicators, not proof"	Covered	Keep; enforce corroboration rule (C-22)
R07	Verify via issuing authorities (APIs, portals, signed records, QR, reference numbers); **record source, timestamp, result**; six statuses; route to manual when unavailable/inconclusive	Covered (simulated)	Add verification **tiers** and feasibility matrix (U-06); store `source`, `queried_at`, `result`, `method` explicitly
R08	Configurable checklists derived from APH, regulations, "authorized instructions"	Covered	Layered rule packs for addenda/instructions (U-01)
R09	Review infrastructure evidence: **photographs, videos, floor plans**, certificates; CV where feasible	Partial — **floor plans not handled**	Add floor-plan area reconciliation (U-09)
R10	Dashboard: application-wise status, deficiencies, verification results, unresolved discrepancies, cases needing intervention	Covered	Keep; keep light (overlaps PS 26262)
R11	Alerts: missing info, expired docs, pending source verification, anomalies	Covered	Keep
R12	Audit trail of **rules applied, records checked, source responses, alerts, observations, actions**	Mostly	Add `rules_applied[]` and `records_checked[]` to each evaluation record (C-26)
R13	Role-based access, secure document handling, confidentiality, **human review before any adverse action**	Partial	Server-side roles; encryption; **state machine that physically blocks LoD issuance without officer confirmation** (C-13, C-15, C-29)
R14	Demo on authorized, anonymized or synthetic data	Covered	Keep; never commit real certificates
R15	Evaluation: extraction accuracy, validation accuracy, anomaly detection, source-verification rate, discrepancy reliability, average verification time, **alert timeliness, reduction in manual effort**	Covered, but spec says "7 metrics"	The brief lists **eight** items; split metric 7 into 7a/7b (C-28)
R16	All checks and workflows configurable	Partial	Rounding policy, thresholds, relaxation rules → config
R17	Do not bypass access controls or assume access to restricted databases	Covered	Never label simulated sources "live" (D-08)
R18	Data protection: security, confidentiality, **access logging, data-retention safeguards**	Partial (**no retention, no encryption**)	C-15, C-16
R19	Explainable, evidence-supported flags	Covered	Keep
R20	AI = decision support; final decisions with authorities	Covered	Keep banner + enforced state machine
R21	"Integrated with, or designed for integration with" the AICTE portal	Production-path table only	Publish OpenAPI + adapter contract + SSO placeholder (Section 5)
R22	Reduce repeated manual checking using records already available	Covered	Keep; add claim-coverage view (U-05)

2.2 Context that shapes the right design

- **AICTE already runs e-governance for approvals** since January 2010 **[V]** (S18). The portal produces deficiency reports **based on self-disclosure [V]** (S19). The gap is *authenticity and cross-checking*, not data entry.
- **The handbook is multi-year but not static.** APH 2024-27 is a three-year handbook **[V]** (S17); AICTE's 2025-26 page links an "Addendum/Corrigendum 2025-26 of APH 2024-27" **[V]** (S25); a 2026-27 addendum also exists **[R]** (S14). Rules therefore need **effective-cycle layering**.
- **2025-26 addendum (reported):** "Adjunct Faculty" (part-time, up to 15% of required faculty) **done away with**; "Professor of Practice" up to **20%** of required faculty (cadre) — 5% exclusively reserved for women PoPs on regular basis **[R]** (S13). The original APH 2024-27 text allowed adjunct *or* PoP up to 15% with special wording for Design and Planning **[V]** (S15, partial text).
- **2026-27 addendum (reported):** supernumerary seats up to 25% for management/design and other technical programmes; institutions required to show "**nil deficiency based on self disclosure**" and submit student details **[R]** (S14).

- **Mandatory website disclosure (Annexure-18):** AICTE's own 2026-27 EoA letter says institutions must disclose, at a prominent place on their website, (i) department-wise infrastructure with approved courses/intake, (ii) faculty details with qualification, tenure, experience, **Student:Faculty Ratio and Cadre Ratio**, (iii) audited financial statements for the last 3 years **[V]** (S12). The same letter cites **Annexure-18 at page 180 [V]** — this conflicts with page numbers in your v2.0.0 (see C-06).
- **Ghost faculty is a recognised problem.** The Education Ministry has approved *voluntary* Aadhaar-based authentication for AICTE-regulated institutions, explicitly to curb ghost faculty **[V]** (S20). Voluntary ⇒ you cannot assume it; your PAN/HMAC fallback is justified.
- **Real consent-based issuer access exists** — DigiLocker Requester APIs (list issued documents, fetch file/XML with integrity HMAC) **[V]** (S21). This is the legitimate route to "verify at the issuer" without scraping.
- **Demand signal:** other institutions are starting to build AI document verification after finding manipulated documents (e.g., Delhi University announced plans in July 2026) **[V]** (S22).
- **Idea-stage competition on this PS is thin** (25 ideas as of 23 Oct per a mirror) **[R]** (S1/S2), but several complete public implementations exist, so finale-stage competition is not thin.
- **An earlier AICTE approval-portal PS on sih.gov.in** asked for evaluator matching, real-time tracking and "digital dimension tracking of infrastructure" **[V]** (S24). The same themes (workflow, dimension checking) keep recurring, which supports adding floor-plan/dimension evidence (U-09).

2.3 Existing solutions — teardown

> All statements are "per README/public page" unless marked. Drawbacks are things you can legitimately raise in a "why yours?" answer — phrase them as *gaps*, never as insults.

2.3.1 Pramana (Anexus5919/pramana) — strongest public competitor (S3)

What it claims (README): 472 commits; 162 passing tests; 29 cited rules built from the APH 2024-27 *and its 2025-26 and 2026-27 addenda*; PDF digital signatures checked against **CCA India root certificates** (offline); issuer-signed QR codes with a public scan-to-verify page; **C2PA Content Credentials** on photos; text-changed-in-later-revision detection with a "time machine"; font and seal-reuse checks; cross-application linking (shared faculty IDs, certificate numbers, file hashes, photos, seals) drawn as a network; year-over-year comparison; deficiency-letter drafting; signed evidence pack; "Ask why" (answers only from its own findings); re-verification watch; inspection brief PDF; pre-submission readiness check; Hindi (Devanagari) OCR; fine-tuned Qwen3-1.7B extractor (94.7% fields fully correct on held-out layouts); parser sandbox; per-document envelope encryption; retention policy; PostgreSQL support; MIT licence; avoids AGPL libraries.

Measured results it reports: 54/54 planted problems on 11 test applications (83 documents); 300 randomised generated applications → 99.2% recall / 91.6% precision before fixes, 100%/100% on two fresh seeds after fixes; 18/18 everyday-editor edit methods; 9/9 real DSC-signed Government e-Gazettes verified.

Gaps you can legitimately raise:

#	Gap (per README)	Our answer
1	Issuer registries are **simulated** (clearly labelled) — real feasibility of each registry not mapped	Verification-tier + feasibility matrix, DigiLocker consent-pull path (U-06)
2	Rule governance beyond "versioned, cited ruleset" and replayable runs is not described: no what-if, impact analysis on pending applications, or maker-checker publish	Layered addenda + impact simulator + maker-checker (U-01, U-02)
3	Reports deficiencies and drafts letters citing clauses, but no **quantified remediation options** or applicant planning tool is described	Remediation engine + Intake Planner (U-03, U-04)
4	No check of the institution's **public mandatory disclosures** is described	Disclosure Auditor (U-07)
5	Evaluation is on its own generated applications (self-graded); real-document test is e-Gazettes, not AICTE-style certificates	Template-shift split, CIs, FAR, red-team matrix (Section 6) — and be honest that you share the synthetic-data limitation
6	**[inference]** Very large feature surface (graph, time machine, Ask-why, evidence packs) plus local-LLM and fine-tuned-model setup raises officer training and operational burden	Governed minimal workflow; deterministic mode by default; LLM optional (U-12/P-05)
7	**[inference]** First-run needs network downloads for model weights/assets	One-command offline-capable demo

What to copy (parity items): CCA-root DSC verification (P-01), PDF revision diff (P-02), encryption + retention + sandbox (P-03/P-04), Hindi OCR (P-05).

2.3.2 CrashCoders (viraj-1709/aicte) (S4)

What it claims: 6-step pipeline; ELA tamper heatmap; SHA-256 duplicate detection across institutions; metadata forensics (Photoshop/Canva/GIMP signatures); chronological sanity; reused identifiers; EXIF GPS Haversine; blueprint room/classroom detection; **transparent 0–100 risk score**; officer confirm/false-alarm feedback that recalibrates scores; hash-chained ledger; PDF dossier; Hindi/English toggle; "Tamper Demo Mode" (clean vs tampered twin); applicant pre-check. Evaluation: 30 synthetic applications; 98.2% extraction, 93.5% time saved (1.2 min vs 18.5 min baseline). Registries simulated via a mock service.

Gaps:

#	Gap	Our answer
1	Forensic signals are individually weak: metadata *software markers* are trivially evaded (print-to-PDF, screenshots) and ELA alone is unreliable on re-compressed scans; exact SHA-256 misses cropped/re-encoded duplicates	Calibrated severity + **corroboration rule** (no single weak signal → Mismatch) + pHash/ORB for near-duplicates (C-22/C-23)
2	Hand-weighted risk score; officer feedback "immediately recalibrates" — ungoverned feedback can drift or be gamed	Feedback goes to a **calibration queue** under maker-checker, never silently changes rules (U-11)
3	Time-saved metric rests on an assumed manual baseline	Time-and-motion protocol with officer volunteers; label as estimate until measured (Section 6.6)
4	Rules appear tied to one handbook snapshot	Layered rule packs (U-01)
5	Single commit, evaluation on own 30 apps	Same limitation applies to you — differentiate on methodology, not on bigger numbers

What to copy: the **Tamper Demo Mode** idea (clean vs tampered twin side-by-side) is excellent for a 3-minute judge demo; a Hindi/English toggle is cheap.

2.3.3 AICTE SmartValidate (sathvik210508) (S5)

What it claims: FastAPI + SQLAlchemy/SQLite + React 19; 7 pre-seeded scenarios (A–G) incl. missing document, expired certificate, faculty-count mismatch (48 declared vs 42 in roster), SHA-256 duplicate across slots, authority mismatch (360 vs 240 intake), low-OCR + source-unavailable (HTTP 503) → manual review; deterministic rule engine with configurable rules; PBKDF2 passwords + HMAC bearer tokens; append-only audit; admin rule manager; analytics dashboards. Its own scope table says authority verification is an adapter returning **synthetic payloads** and does not call real databases.

Gaps: forensics limited to SHA-256 duplicates and chronology (no QR/DSC/font/ELA); demo is scenario-driven; no programme-wise FSR/cadre depth visible; no addenda handling.
What to copy: the **low-confidence OCR + source-unavailable ⇒ route to manual, not accuse** scenario — it is exactly the safe-failure behaviour the brief wants.

2.3.4 SIH11 (sohamramshette/SIH11) (S6)

What it claims: React 19 + Vite + Tailwind front-end portal with four role consoles (Institution, Verification Officer, Senior/Directorate Officer, Admin), public portal with application-status search, rule-engine configuration UI, "AI vs human review" distinction, officer performance metrics, CSV export, cryptographic audit trail.

Gaps: repository (as fetched) contains only the front-end folder; role-based access is a "simulation"; no back-end, forensics or evaluation evidence in the repo; 3 commits.
What to copy: a **Directorate/Senior-officer approval step** for adverse actions → natural home for maker-checker.

2.3.5 CipherCore Verify (nonutsayan.github.io/CipherCore-verify) (S7)

What it claims: single-page demo; "direct API validation against authoritative government registries (State Fire, DigiLocker State Land Registries, University Affiliation)"; "YOLOv8x-SIH26261" workstation counting; "SHA-256 Merkle chaining" ledger and a "7-validator consortium PoA network (AICTE HQ, IIT Bombay, …)"; six mandatory documents attributed to "APH Section 2.1"; norms such as "FSR 1:20" and "terminal density 1:1 per intake".

Gaps:

- Presents mock data as "live government API" — conflicts with the brief's rule *not to assume access to restricted databases* (R17).
- Blockchain consortium and fine-tuned detector appear to be **client-side simulations** — over-claiming that a technical judge will probe.
- Its six-document list does not match AICTE's actual portal document list (which centres on Affidavit-2, occupancy/completion certificate, structural stability certificate for >30-year-old buildings, fire safety certificate, etc.) **[V]** (S11).
- Infrastructure norms stated without clause references.
  Lesson: over-claiming is the fastest way to lose credibility — see the Claims Register (Section 7.6).

2.3.6 Non-hackathon alternatives

Category	Examples	What they do	Why they are not a substitute
AICTE's own portal	e-governance approval system (since 2010)	Self-disclosure-based deficiency reports; committees (Scrutiny, Re-Scrutiny, EVC, SAC/SHC)	Does not authenticate documents or cross-check across institutions/websites
DigiLocker / API Setu	Requester APIs; issued-document fetch with integrity HMAC	Consent-based retrieval of issuer-signed documents	Plumbing, not scrutiny; only issuers integrated; user consent needed
KYC / document-verification vendors	Generic "AI document verification" APIs (e.g., for admissions mark-sheets, ID cards)	Identity/marksheet classification and checks	No AICTE rulebook, no institutional infrastructure/faculty logic, no EVC/LoD workflow

Category	Examples	What they do	Why they are not a substitute
Other regulators' tools	University admission-scrutiny efforts (DU plans)	Detect manipulated admission documents	Different document population and rules

2.4 Cross-cutting drawbacks of existing solutions (and our counter)

ID	Drawback seen across the field	How v2.1 answers it
D-01	**Frozen rules** — one snapshot of the handbook; no addenda/cycle layering; no impact analysis	U-01 layered rule packs, U-02 impact simulator
D-02	**Simulated registries with no feasibility story** (some labelled "live")	U-06 verification tiers + feasibility matrix; DigiLocker consent-pull; offline-verifiable signals first
D-03	**Self-graded evaluation** on the team's own generator; headline "0 false alarms"; no CIs	Section 6: template shift, Wilson CIs, FAR, abstention, red-team matrix; honest synthetic caveat
D-04	**Detection without remedy**	U-03 remediation engine, U-04 Intake Planner
D-05	**No view of "declared vs proven"** and no check of the public website disclosures AICTE mandates	U-05 claim coverage, U-07 Disclosure Auditor
D-06	**Fragile single-signal forensics** (software-name metadata, ELA alone, exact hash)	Corroboration rule; calibrated severity; pHash+ORB; revision diff (P-02)
D-07	**Security/privacy gaps** (quick-login demo roles, no encryption/retention, unsandboxed parsing)	C-13…C-18, P-03, P-04
D-08	**Over-claiming** (live APIs, consortium blockchain, model names)	Claims Register; honest labels; "Simulated" watermark on adapter responses
D-09	**Ungoverned feedback loops and no consistency view**	U-11 governed calibration; U-10 consistency index
D-10	**Operational friction** (heavy asset downloads, model training before demo)	One-command Docker; deterministic default; LLM optional
D-11	**Floor-plan/dimension evidence barely addressed**, despite the brief naming floor plans	U-09 floor-plan reconciliation
D-12	**Regional-language handling is UI-only** in most	P-05 multilingual OCR with confidence gating

2.5 Capability matrix (README-level; "not seen" ≠ "absent")

Capability	v2.0.0	v2.1	Pramana	CrashCoders	SmartValidate	SIH11	CipherCore
Category-scoped checklist	✔	✔	✔	✔	✔	UI	✔ (list differs)
Cited, versioned rule pack	✔	✔	✔	partial	partial	UI	✘
Base + addenda overlays with effective cycle	✘	✔ **U-01**	not seen	✘	✘	✘	✘
Impact simulation before publish	partial	✔ **U-02**	not seen	✘	✘	✘	✘
Quantified remediation / planner	✘	✔ **U-03/04**	not seen	✘	✘	✘	✘
Public-website disclosure audit	✘ (boolean input)	✔ **U-07**	not seen	✘	✘	✘	✘
Verification tiers + feasibility matrix	✘	✔ **U-06**	partial (proof ladder)	✘	✘	✘	✘
Real DSC check vs CCA roots	prod path	✔ **P-01**	✔	✘	✘	✘	claim only
PDF revision-level diff	✘	✔ **P-02**	✔	metadata	✘	✘	partial

Capability	v2.0.0	v2.1	Pramana	CrashCoders	SmartValidate	SIH11	CipherCore
QR payload vs visible text	✔	✔	✔	not seen	✘	✘	claim
ELA (as supporting evidence)	✔	✔ (calibrated)	not seen	✔	✘	✘	✔
Near-duplicate photo (pHash+ORB)	✔	✔	fingerprints	exact hash	✘	✘	✘
Cross-institution faculty dedup	✔	✔	✔ (+graph)	partial	✘	UI	partial
Floor-plan area reconciliation	✘	✔ **U-09**	not seen	partial	✘	✘	partial
Encryption at rest + retention	✘	✔ **P-03**	✔	partial	✘	✘	✘
Hindi OCR	partial (config)	✔ **P-05**	✔	UI only	✘	✘	✘
Measured eval w/ CIs, FAR	targets	✔ **§6**	strong (self-graded)	small	✘	✘	✘
One-command offline demo	✘	✔	needs asset fetch	✔	✔	✔	✔

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3. Corrections to spec v2.0.0

3.1 Index (severity: **Crit** = credibility or security failure if left; **High**; **Med**; **Low**)

ID	Area	Correction	Sev	Effort
C-01	Regulatory	Adjunct/PoP caps are outdated (2025-26 addendum)	Crit	S
C-02	Regulatory	Cadre formula: Principal offset, rounding policy	Crit	S
C-03	Regulatory	Planning/Architecture cadre (1:2:6 vs 1:3:10)	High	S
C-04	Regulatory	Unverified FSR rows; Diploma formula missing from evaluator	High	S
C-05	Regulatory	Low-enrollment relaxation: exact condition and scope	High	S
C-06	Regulatory	Page/annexure references conflict; add `verified` flag	Crit	S
C-07	Regulatory	Affidavit-2 is portal-generated; validate against application record	Med	S
C-08	Regulatory	Structural certificate: issuer wording, validity boundary	High	S
C-09	Model	"Missing" is not a verification status — add completeness dimension	High	S
C-10	Model	Rule pack must model addenda and cycle (→ U-01)	Crit	M
C-11	Regulatory	Annexure-4: PC:student ratio table missing; numbers unverified	High	S
C-12	Regulatory	Annexure-14 covers 5 states only	Med	S
C-13	Security	Role chosen by the client at login	Crit	S
C-14	Privacy	Raw PAN stored; Aadhaar-derived values	Crit	S
C-15	Security	No encryption at rest; page PNGs served unauthenticated	Crit	M
C-16	Privacy	No retention policy (brief requires it)	High	S
C-17	Security	Hash chain is bypassable without external anchoring	High	S
C-18	Security	Untrusted-file parsing, upload hardening, LLM injection	High	M
C-19	Licensing	AGPL dependencies (PyMuPDF, Ultralytics)	High	M
C-20	Build	`requirements.txt` gaps and mismatches	Med	S
C-21	Logic	`/365.25` date arithmetic gives wrong answers on anniversaries	Crit	S
C-22	Forensics	Corroboration rule — no single weak signal → Mismatch	Crit	S

ID	Area	Correction	Sev	Effort
C-23	Forensics	ELA limitations; use as supporting evidence only	High	S
C-24	Forensics	Font-subset check false-positive control	Med	S
C-25	Forensics	Photo-reuse thresholds and same-Trust handling	Med	S
C-26	Audit	Record rules applied, records checked, hashes of inputs	High	S
C-27	Data	Missing history tables required by the brief	High	M
C-28	Evaluation	Brief lists **eight** metrics; benchmark values are targets	Crit	S
C-29	Workflow	Enforce human review before adverse action in code	High	M
C-30	Privacy	Faculty identity path vs AICTE's Aadhaar authentication	Med	S
C-31	Model	Status roll-up semantics and manual-confirmation path	Med	S
C-32	Scope	Scope drift to PS 26262; illustrative formats presented as facts	Med	S

C-01 Adjunct / Professor-of-Practice caps are outdated — Crit

Issue. v2.0.0 hard-codes `adjunct_and_pop_max_pct = {Design: 20, Planning: 30, DEFAULT: 15}` and a rule `ADJUNCT_POP_AND_PG_PHD`.

Evidence.

- Original APH 2024-27 text: "Adjunct Faculty **or** Professor of Practice up to 15% of the required faculty strength", with separate wording for Design and Planning courses **[V]** (S15, partial text — the Design/Planning figures are truncated in the copy I read, so your 20/30 are **[?]**).
- 2025-26 addendum (reported): Adjunct Faculty category **done away with**; PoP (including Associate/Assistant PoP) up to **20%** of required faculty strength (cadre), **5%** exclusively reserved for women PoPs on regular basis **[R]** (S13).
- UGC's separate guideline caps PoP at 10% of sanctioned posts — a different regulator; do not mix **[V]** (S13).

Fix. Never store this as a flat constant. Model it as a **base value + dated overlay** (U-01):

```
// base/2024-25/faculty_norms.json
{
  "adjunct_or_pop": {
    "default_max_pct_of_required_faculty": 15.0,
    "Design":   {"max_pct": null, "note": "wording exists in APH 2024-27; value [?] verify"},
    "Planning": {"max_pct": null, "note": "wording exists in APH 2024-27; value [?] verify"},
    "source_ref": {"doc": "APH 2024-27", "clause": "[?]", "page": "[?]", "verified": false}
  }
}

// overlays/addendum_2025_26.json
{
  "overlay_id": "ADD_2025_26_POP",
  "applies_from_cycle": "2025-26",
  "source_ref": {"doc": "Addendum/Corrigendum 2025-26 of APH 2024-27", "clause": "[?]", "page": "[?]", "verified": false},
  "ops": [
    {"op": "remove", "path": "/adjunct_or_pop"},
    {"op": "set", "path": "/professor_of_practice", "value": {
        "max_pct_of_required_faculty": 20.0,
        "reserved_for_women_pct": 5.0,
        "women_reserved_basis": "regular PoP / Associate PoP / Assistant PoP",
        "counts_against_cadre": true,
        "scope_note": "reported for engineering and technology; confirm other streams"}}
  ]
}
```

Test. UG Engineering, required faculty 60, PoP engaged 14 → under 2024-25 rules: 14/60 = 23.3% > 15% (flag); under 2025-26 overlay: 23.3% > 20% (flag); PoP 11 → 18.3%: flag under 2024-25, pass under 2025-26. This single case is a perfect live demo of *why layering matters* (Section 7.4).

C-02 Cadre formula: Principal offset and rounding — Crit

Issue. v2.0.0 states `Prof = LF_req / RJ × 1`, `Assoc = LF_req / RJ × c2`, `Asst = F_req − Prof − Assoc`. It (a) floors, (b) ignores that the Principal/Director is counted separately, (c) leaves the rounding policy implicit.

Evidence. The published UG table (E&T) is **[V]** (S8, S9): Principal/Director = 1; Professor = `S/(20×R) − 1`; Associate = `S/(20×R) × 2`; Assistant = `S/(20×R) × 6`; Total = `S/20`; `R = 1+2+6 = 9`. PG E&T uses `R = 1+1+1` with no Principal row; MCA/MBA/PGDM use `R = 9` with Principal 1.

Fix.

```
import math
EPS = 1e-9
def ceil_tol(x: float) -> int:          # avoids 6.0000000001 -> 7
    return math.ceil(x - EPS)

def ug_cadre(S, fsr, ratio=(1, 2, 6), principal=1, relax=False, policy="ceil"):
    R = sum(ratio)
    total = (S / fsr) * (0.75 if relax else 1.0)
    unit = total / R
    rnd = ceil_tol if policy == "ceil" else round      # policy is config [A]
    prof_cadre = rnd(unit * ratio[0])                  # includes Principal/Director
    return {
        "total_required": rnd(total),
        "professor_cadre_incl_principal": prof_cadre,
        "professors_excl_principal": max(prof_cadre - principal, 0),
        "associate": rnd(unit * ratio[1]),
        "assistant": rnd(unit * ratio[2]),
    }
```

Worked example (use as unit test). UG Engineering, `S = 1080`, `fsr = 20`:

- `S/fsr = 54`, `unit = 54/9 = 6` → Professor cadre 6 (Principal 1 + 5 Professors), Associate 12, Assistant 36, total 54 ✔.
- With relaxation (×0.75): total 40.5 → 41; unit 4.5 → Professor cadre 5, Associate 9, Assistant 27 (sum 41 ✔).

Rounding is [A]. The handbook table shows fractions; officers round in practice. Expose `rounding_policy` in `rules.json` and print it in every finding ("computed with ceiling rounding, tolerance 1e-9").

C-03 Planning/Architecture cadre — High

Issue. v2.0.0 uses `[1,2,6]`, `R=9` for UG Planning.

Evidence. AICTE's formula table uses `R=9` for UG Planning (FSR 1:16) **[V]** (S8). The same source lists a **minimum cadre of 1:3:10** under CoA-based norms for standalone Architecture and Town Planning **[V]** (S8, section 6.3), plus special rules for allied-area faculty and tenure faculty (S8, note 6).

Fix. Make cadre a property of `(course_level, programme_stream, institution_kind)`; add `institution_kind ∈ {STANDARD, STANDALONE_ARCH_PLANNING}` with its own ratio and allied-faculty caps; mark which clause governs. When unsure, route to **Manual Review Required** rather than computing a deficiency.

C-04 Unverified FSR rows and missing Diploma formula — High

Verified rows [V] (S8, S9): UG E&T 1:20; UG Planning 1:16; UG Applied Arts & Crafts 1:15; UG Design 1:15; UG HMCT 1:20; PG E&T 1:15 (`R=3`); PG Planning 1:10 (`R=3`); MCA 1:20 and MBA/PGDM 1:20 (`R=9`, Principal 1).

Unverified [?]: UG Computer Applications 1:25, UG Management 1:25 (BBA/BCA/BMS came under AICTE from 2024-25 **[V]** S17 — the ratios themselves need the APH Appendix); PG Design, Applied Arts & Crafts, HMCT ratios (15/15/12 in v2.0.0).

Diploma [V] (S10, older handbook; confirm for 2024-27): Principal 1; HoD 1 per department; Lecturers = `S/25 − (Principal + HoD)`; total `S/25`. Your matrix stores `hod_per_dept` but the evaluator text has no formula — add it.

Fix. Tag each cell `verified: true|false`. Unverified cells evaluate to **Manual Review Required**, never to a deficiency.

C-05 Low-enrollment relaxation — High

Evidence [V] (S8): "If the average admission during the last 3 years is ≤ 50% of the average sanctioned intake, the requirement of faculty may be reduced by 25%" (worked example in the source multiplies 60 by 0.75 = 45).

Fix.

- Input is `avg_admissions_3yr / avg_sanctioned_3yr ≤ 0.50`. Store both numerators/denominators, not just a percentage.
- Apply at **programme** level; surface "relaxation applied" as a finding with the three yearly figures.
- Make the *scope* configurable `relaxation_applies_to: [...]` because the handbook example may be limited by programme type **[?]**.
- Edge: any of the three years missing → relaxation **not** applied and finding says why.

C-06 Page and annexure references conflict — Crit

Issue. v2.0.0 cites Annexure-14 at p. 180 and Annexure-18 at p. 184. AICTE's own 2026-27 EoA letter cites "**Annexure-18, page 180" [V]** (S12). At least one of your references is wrong, which undermines the "every rule has a verifiable `source_ref` " claim.

Fix.

1. Replace every `page` value that you have not personally checked in the PDF with `"page": null` .
2. Add `verified: bool` , `verified_by` , `verified_on` , `pdf_sha256` .
3. UI shows an amber "citation not yet verified" chip for `verified:false` .
4. Pair every clause number with **text anchor** (first six words of the clause) so the reference survives page re-numbering across versions.

```
"source_ref": {
  "doc": "APH 2024-27", "pdf_sha256": "<hash of the file you used>",
  "chapter": "VI", "clause": "(h)", "anchor": "Structural Stability Certificate is required if",
  "page": 80, "verified": true, "verified_by": "<initials>", "verified_on": "2026-10-26"
}
```

C-07 Affidavit-2 handling — Med

Evidence. The portal-generated affidavits carry the **application number, institute type, region and date** in the header and are generated from the application itself **[V]** (S16; S11 lists "Portal Generated Affidavit 2"). The "sworn before First Class Judicial Magistrate / Notary Public / Oath Commissioner" requirement is reported in handbook text **[R]** (S26). The ₹100 stamp minimum in v2.0.0 is **[?]**.

Fix. Validate the affidavit **against the application record**: application number, institution type, state/region, date within the cycle, and that its declarations (e.g., "uploaded occupancy/completion certificate and structural stability certificate") are *consistent with uploaded documents* — a declared-but-missing document is a strong, explainable finding. Keep the DSC/stamp checks, but mark stamp amount `[?]` until verified.

C-08 Structural stability certificate — High

Evidence. AICTE's portal document list says "**Stability Certificate from Competent Authority in case existing buildings are of more than 30 years old" [V]** (S11). v2.0.0 says "registered Structural Engineer" and cites a certificate format page.

Fix.

- Accept both `issuer_type ∈ {COMPETENT_AUTHORITY, REGISTERED_STRUCTURAL_ENGINEER}` **only if** the clause text allows it; otherwise route a mismatch to Manual Review (not a deficiency) [?].
- The "valid 5 years" and "Government buildings > 50 years" exception are **[?]** — verify.
- Trigger uses calendar arithmetic (C-21), with an explicit `boundary` policy (inclusive/exclusive) and a "±1 day near-boundary ⇒ Manual Review" guard.

C-09 "Missing" is not a verification status — High

Issue. v2.0.0 reports a missing mandatory conditional document as `Mismatch Detected` . The brief's six statuses describe **verification outcomes**; "missing/incomplete" is a **completeness** finding.

Fix. Two orthogonal dimensions per requirement:

Dimension	Values
`completeness`	`COMPLETE` , `MISSING` , `INCOMPLETE` , `NOT_APPLICABLE`
`verification_status`	the six brief statuses

Roll-up shows both ("3 missing, 2 mismatches"). A missing document has `verification_status = null` (nothing to verify), not "Mismatch".

C-10 Rule pack must model addenda and cycle — Crit (implemented as U-01)

v2.0.0's `manifest.json` ("AICTE_APH_2024_2027, rule_pack_version 2.0.0, effective 2026-04-01 … 2027-03-31") treats the three-year handbook as one frozen artifact. Replace with **base + overlays + cycle resolution** (Section 4, U-01) and add a `cycle` parameter to `evaluate()` :

```
def evaluate(facts, rule_stack, cycle: str, as_of_date: str, what_if_overrides=None) -> dict: ...
```

`cycle` selects *which rules apply*; `as_of_date` selects *how document validity is computed*. They are different.

C-11 Annexure-4 gaps — High

- `DEPT_PC_AND_RATIO_CHECK` says "check PC:Student ratio per Annexure-4 (1:6 to 1:10)" but `rules.json` has **no ratio table**, only `min_pcs_per_department_lab = 20`.
- Numbers `20 PCs`, `5%` printers, `1` A1 plotter, bandwidth tiers `100/300/500/1000 Mbps` are **[?]**.
- Competitors state norms like "1:1 terminals per intake" without clauses (S7) — do not copy.

Fix. Add the ratio table **only after** you copy it from the PDF; until then the rule yields `Manual Review Required: "ratio table not loaded"`. Mark each constant `verified`.

C-12 Annexure-14 fire authorities — Med

Five states are not enough, and an unknown state must not become an accusation.

Fix. `state_authority_registry` with `status ∈ {VERIFIED_LIST, PARTIAL, UNKNOWN}`; unknown state or unlisted authority name ⇒ **Manual Review Required** with message "issuing authority not in loaded Annexure-14 list (state X)". Add fuzzy name matching (rapidfuzz) and a manual-add workflow for officers (maker-checker).

C-13 Role chosen by the client at login — Crit

Issue. `POST /api/auth/login {"username": "...", "role": "Scrutiny_Officer"}` lets any caller request any role.

Fix.

```
users(user_id, username, pw_hash_argon2id, role, region, is_active, totp_secret_enc, created_at)
POST /api/auth/login {username, password, totp?}  -> JWT {sub, role(from DB), region, jti, exp<=60min}
Refresh token rotation; revoke list on logout; lockout after N failures; rate-limit per IP/user.
Object-level checks: Applicant sees only own inst_id; officers see only their region; Rules_Admin cannot decide applications.
```

The demo "role switcher" becomes **separate seeded logins** or an `Admin impersonate` action that writes an audit entry. In production map identities from AICTE SSO/NIC (placeholder adapter).

C-14 PAN and Aadhaar-derived values — Crit

Issue. You promise "zero raw government ID storage" yet store raw `pan_number` for ghost-faculty dedup.

Fix.

- Store `pan_digest = HMAC-SHA256(K_pan, normalized_pan)` and `pan_last4` for display; `K_pan` from env/KMS, **not** in the DB; plan dual-digest rotation.
- Dedup joins on digest only.
- **Aadhaar**: AICTE has been permitted *voluntary* Aadhaar authentication **[V]** (S20). Do not persist any Aadhaar-derived value (including digests) unless legal review says it is allowed **[?]**; consume only the *verification result + reference id* from AICTE's own service in production.
- Redaction tests: grep logs/PDFs/JSON for 10-char PAN and 12-digit patterns in CI.

C-15 Encryption at rest; unauthenticated page images — Crit

Issues. (1) Documents and rendered PNGs are stored in clear. (2) "All endpoints except `/api/auth/login` and `/static/*` require JWT" means page PNGs are **publicly fetchable**.

Fix.

- **Envelope encryption:** per-document random 256-bit DEK, AES-256-GCM; DEK wrapped by KEK (env now, KMS/HSM in production); store `enc_alg`, `wrapped_dek`, `nonce`, `key_id`.
- PNG pages: generate on demand or store encrypted; serve through **authenticated** `GET /api/pages/{doc_id}/{n}` or short-lived (≤60 s) signed URLs bound to user and doc.
- Never log extracted PII values; log field names + hashes.

C-16 Retention — High

The brief explicitly requires "data-retention safeguards" **[V]** (S1).

Fix. `retention_policy` config (e.g., documents N days after application closure/withdrawal), `legal_hold` flag, nightly purge job that deletes blobs and derived images but keeps **hash + metadata tombstone** so the audit chain stays valid; each purge writes an audit entry. Align wording with the DPDP Act 2023 principles (purpose limitation, storage limitation) **[?]** — have someone with legal context review.

C-17 Audit chain anchoring — High

SQLite triggers stop casual edits but anyone with file access can drop triggers and recompute the chain. It is *tamper-evident only if the head is held elsewhere*.

Fix.

- `chain_checkpoints(log_id, head_hash, mac, created_at)` where `mac = HMAC(K_audit, head_hash || log_id || ts)`; create a checkpoint every N entries.
- Publish the head hash to a second location (separate store / signed git commit / RFC 3161 timestamp if a TSA is available).
- `verify_chain()` checks: chain continuity, checkpoint MACs, and that the externally published head is a prefix of the local chain.
- State the property honestly: "tamper-evident, not tamper-proof"; production path = PostgreSQL with append-only role + WORM replication.

C-18 Hostile-input handling — High

Fix.

- Parse PDFs/images in a **sandboxed subprocess** (rlimit CPU/memory, wall-clock timeout) — Pramana does this (S3), so judges may ask.
- Reject: encrypted PDFs, PDFs with JavaScript/Launch actions/embedded files, > N pages, > M MB, image decompression bombs (`PIL.Image.MAX_IMAGE_PIXELS`).
- Sniff true file type; random storage ids; never use user filenames for paths.
- If an LLM extractor is added (P-05): treat document text as **data**; constrain output to a JSON schema; accept an extracted value only if it is **found verbatim/normalised in the OCR text** (grounding check); never act on instructions found in documents; add a "planted instruction" test set (Pramana reports 0/30 followed — S3).

C-19 Licensing — High

Component in v2.0.0	Issue	v2.1 replacement
PyMuPDF (`fitz`)	AGPL-3.0 / commercial dual licence **[?]** (confirm on project page)	`pypdfium2` (render), `pdfplumber` / `pdfminer.six` (text, boxes), `pikepdf` (structure, revisions), `pypdf`
Ultralytics YOLOv8	AGPL-3.0 / enterprise licence **[?]**	Apache-2.0 detector weights via ONNX Runtime (e.g., YOLOX); confirm each weight's licence
PaddleOCR (named, not in requirements)	Apache-2.0 but heavy	`rapidocr-onnxruntime` (Apache-2.0) + Tesseract fallback

Pramana's README independently says it avoids AGPL PDF libraries and AGPL/non-commercial detector/OCR weights **[V]** (S3). A government deployment path with copyleft network clauses is a legitimate jury question.

C-20 `requirements.txt` gaps — Med

Missing or mismatched: PaddleOCR (named in stack, absent from file); Tesseract binary and `eng+hin+tam` language packs (need `apt`); `python-dateutil`; `argon2-cffi`; `cryptography`; `rapidfuzz`; `pikepdf`; `pdfplumber`; `onnxruntime`; `rapidocr-onnxruntime`; `httpx`; `selectolax` or `beautifulsoup4`; `pytest`, `hypothesis`. `ultralytics` drags in PyTorch (large) — remove. Provide a `Dockerfile` + `docker-compose.yml` that builds with no network at run time (D-10).

C-21 Date arithmetic — Crit

Issue. `Δ = (as_of − issue) / 365.25`, then `> 30.0` / `≤ 5.0`.

Why it is wrong. The ratio is not calendar-exact. Counter-examples:

Check	Issue date	As-of	Days	Days/365.25	`/365.25` verdict	Correct (calendar)
"more than 30 years"	1996-01-01	2026-01-01	10,958	30.0014	**> 30 → triggers**	exactly 30 years → **does not** trigger
"valid 5 years from issue"	2020-01-01	2025-01-01	1,827	5.0021	**> 5 → expired**	exactly 5 years → boundary policy decides

Fix.

```
from datetime import date, timedelta
```

```
def add_years(d: date, n: int) -> date:
    try:
        return d.replace(year=d.year + n)
    except ValueError:                      # Feb 29 -> Feb 28 policy [A]
        return d.replace(year=d.year + n, day=28)

def older_than(issue: date, as_of: date, years: int) -> bool:
    return as_of > add_years(issue, years)

def valid_for_years(issue: date, as_of: date, years: int, inclusive_end=True) -> bool:
    end = add_years(issue, years) - (timedelta(0) if inclusive_end else timedelta(days=1))
    return as_of <= end

def near_boundary(issue, as_of, years, days=1):   # route to Manual Review
    return abs((as_of - add_years(issue, years)).days) <= days
```

Add property-based tests (`hypothesis`) over random dates; include the table above as fixed unit tests.

C-22 Corroboration rule and severity model — Crit

Principle. The brief says flags are "preliminary indicators requiring human verification, not conclusive proof" **[V]** (S1). A single weak signal must never produce `Mismatch Detected` .

Signal family	Examples	Strength
Cryptographic / issuer	DSC bytes modified after signing; issuer-signed QR payload contradicts visible value; issuer source (T1/T2) contradicts submitted value	**High**
Structural	Later-revision text overlay; value span in font unused elsewhere in doc/template; baseline shift	**Medium**
Pixel	ELA local-variance anomaly at value bbox; splice boundary	**Medium (supporting)**
Semantic / cross-document	Survey number mismatch across deed/plan/form; date-order impossibility	**Medium–High** (High if the contradicting record is authoritative)
Identifier reuse	Same certificate number / seal / photo on another institution	**High** if cross-Trust; Medium if same Trust (C-25)
Hygiene	Scanner producer tag, incremental save, mod-date > creation-date	**Low** — display only

Decision table.

Evidence combination	Resulting status
≥ 1 High	`Mismatch Detected`
≥ 2 Medium from **different families**	`Mismatch Detected`
exactly 1 Medium	`Manual Review Required` (+ explanation, benign alternatives)
Low only	no status change (info chip)
Low OCR confidence / source unavailable / near-boundary date	`Manual Review Required` / `Source Unavailable` — **never accuse**

Every flag stores `benign_explanations[]` (e.g., "scanner re-compression", "font fallback for Devanagari") shown to the officer.

C-23 ELA — High

ELA assumes a single prior JPEG compression; scanned documents are already re-compressed and modern editors re-save images, so ELA produces both false positives and false negatives. **Use it as supporting evidence only** (≤ Medium, only when the anomaly is localised on the value bounding box *and* structural evidence is absent or ambiguous). Report ELA's standalone precision/recall on your hard-negative scans in the evaluation (Section 6) rather than asserting it works.

C-24 Font-subset check — Med

False-positive sources: bold label vs regular value on the same line (normal); font fallback for Devanagari/Tamil glyphs; PDF producers that subset fonts per page.

Fix. Compare a value span's font against **other value spans of the same field type across the document/template**, not against its label. Measure baseline shift relative to the line's median baseline; ignore $|\Delta y_0| < 0.3$ pt (rounding). Require both anomalies (font + shift) for Medium; either alone ⇒ Low.

C-25 Photo reuse thresholds and same-Trust handling — Med

- `pHash ≤ 5` and `ORB inliers ≥ 18` are **starting points [A]**; derive thresholds from an ROC on a dev set containing near-duplicates and unrelated look-alike labs (similar furniture layouts). Report the operating point and its FAR.
- Cross-Trust reuse ⇒ High. **Same-Trust** reuse ⇒ Medium (could be shared campus or generic stock) ⇒ Manual Review. Same institution, previous year ⇒ Low unless the claim is "newly built lab".
- Always show the matched images side by side with keypoint lines (you already do).

C-26 Audit record content — High

The brief wants "rules applied, records checked, source responses, alerts, observations, actions" **[V]** (S1). Add an `evaluation_runs` table and reference it from `audit_trail`:

```
CREATE TABLE evaluation_runs (
  run_id TEXT PRIMARY KEY, inst_id TEXT NOT NULL, cycle TEXT NOT NULL, as_of_date DATE NOT NULL,
  facts_sha256 TEXT NOT NULL, ruleset_sha256 TEXT NOT NULL, evaluator_version TEXT NOT NULL,
  rules_applied_json TEXT NOT NULL,      -- [{rule_id, layer_id, source_ref_hash}]
  records_checked_json TEXT NOT NULL,    -- [{table, key, version}]
  findings_json TEXT NOT NULL, rollup_status TEXT NOT NULL, created_at_ist TEXT NOT NULL
);
```

C-27 Missing history tables — High

Brief R02 lists previous EoA, approved programmes/intake, accreditation, compliance/deficiency status, TER status, application history. Add:

```
CREATE TABLE eoa_history (inst_id TEXT, academic_year TEXT, programme_stream TEXT, course_level TEXT,
  approved_intake INTEGER, status TEXT, letter_ref TEXT, PRIMARY KEY (inst_id, academic_year, programme_stream, course_level));
CREATE TABLE application_history (app_id TEXT PRIMARY KEY, inst_id TEXT, academic_year TEXT, category TEXT,
  submitted_at TEXT, outcome TEXT, lod_count INTEGER);
CREATE TABLE accreditation_records (inst_id TEXT, programme TEXT, body TEXT, valid_from DATE, valid_to DATE, ref_no TEXT);
CREATE TABLE deficiency_history (inst_id TEXT, academic_year TEXT, deficiency_code TEXT, status TEXT, closed_on DATE);
CREATE TABLE ter_ledger (inst_id TEXT, academic_year TEXT, amount_inr INTEGER, status TEXT, receipt_no TEXT);
```

New checks enabled: intake claimed ≠ intake approved last year; NBA/NAAC claim vs record validity; repeated deficiency patterns; TER paid for a *different* year.

C-28 Evaluation framing — Crit

1. The brief lists **eight** items (extraction accuracy, validation accuracy, anomaly detection, source-verification rate, discrepancy-identification reliability, average verification time, alert timeliness, reduction in manual effort) **[V]** (S1). v2.0.0 says "seven". Split metric 7 into **7a alert timeliness** and **7b manual-effort reduction**.
2. The values in the "Target Prototype Benchmark" column (96.8%, 98.0%, 93.8/100/96.8, 100%, 97.8%, 86.1%) are **targets chosen before any run**. Label them "targets"; report measured values beside them.
3. "Successful source-verification rate = handshakes resolved without unhandled exception" is trivially 100% for a simulator. Redefine as **status-mapping accuracy under fault injection** (Section 6.3).
4. The 18-minute manual baseline is **[A]**; replace it with a measured time-and-motion estimate (Section 6.6) or keep the label "assumed".

C-29 Enforce human review in code — High

Add an explicit state machine; the API must make adverse transitions impossible for the system actor.

```
RECEIVED → AUTO_SCREENED → OFFICER_REVIEW → { CLEARED
                                            | LOD_DRAFTED → LOD_CONFIRMED (officer, justification) → LOD_ISSUED
                                            | EVC_RECOMMENDED (officer)
                                            | SPEAKING_ORDER_ESCALATED (regional officer) }
```

Rules: `System_AI` may only reach `AUTO_SCREENED` / `LOD_DRAFTED`; `LOD_ISSUED` requires `Scrutiny_Officer` + justification text; `SPEAKING_ORDER_ESCALATED` requires `Regional_Officer`; `Mark_Verified_Override` on a High finding needs a second approver (maker-checker). Enforce in service layer and with a DB `CHECK`/trigger on allowed transitions.

C-30 Faculty identity path — Med

State the real-world path in the deck: AICTE-side verified faculty identifiers (possibly via voluntary Aadhaar authentication **[V]** S20) are the production source of truth; your PAN-digest dedup is a **fallback and a cross-check**, useful because Aadhaar use is voluntary. Add a data-quality rule: PAN format validation (10 chars, pattern) and duplicate-within-institution detection.

C-31 Status roll-up semantics — Med

Keep your precedence `Mismatch > Expired > Manual Review > Verification Pending > Source Unavailable > Verified`, but define:

- **Requirement-level** statuses roll up to application level; `Verified` only if *all mandatory requirements* are `Verified`, completeness is `COMPLETE`, and no open High flags.
- `Source Unavailable` is **not an institution failure**: retries with backoff + circuit-breaker; after N attempts → officer task "confirm manually".
- Manual confirmation by an officer results in `Verified` with `method = MANUAL_CONFIRMATION`, officer id, evidence note (so the six-status contract is preserved).
- Always display the **count breakdown** beside the roll-up badge.

C-32 Scope drift and "illustrative" formats — Med

- Queue/SLA/undertaking tracker/alert centre overlap with **PS 26262** (workflow automation, compliance monitoring, alerts) **[V]** (S2). The brief for 26261 still requires a dashboard and alerts — keep them *minimal*.
- Formats given as facts but unverified: SHCIL verification URL pattern, ICAI UDIN "18-digit" (UDIN is commonly described as an 18-character alphanumeric code **[?]**), "TER" expansion/Bharatkosh behaviour. Label every such item "illustrative format — simulated adapter" in the deck and UI.
- YOLO workstation counting: keep **advisory** and optional; AGPL issue (C-19). EVC walkthrough ≥ 30 min check is cheap (`ffprobe` duration) but its clause **[?]** must be verified.

---

4. New unique additions

4.1 Prioritisation

"Overlap" = whether any public competitor README I read shows something similar (not proof of absence).

ID	Feature	Judge-visible impact	Overlap with public repos	Effort	Tier
U-01	Layered rule packs: base handbook + dated addenda, cycle resolution, per-value provenance	Very high	None seen	M	**A**
U-02	Impact simulator + maker-checker publish gate	Very high	None seen	M	**A**
U-03	Counterfactual remediation engine ("what would make this pass")	Very high	None seen	M	**A**
U-06	Verification tiers + feasibility matrix + DigiLocker consent-pull path	High	Partial (Pramana proof ladder)	S–M	**A**
U-12	Safety panel: False-Accusation Rate, abstention, CIs	High	None seen (others report "0 false alarms")	S	**A**
U-04	Intake Planner for applicants (inverse of the evaluator)	High	Partial (Pramana readiness check)	M	B
U-07	Disclosure Auditor: public-website three-way check	High	None seen	M–L	B
U-05	Claim coverage ("declared vs proven") + risk-based inspection focus	Medium–High	Partial (proof ladder, inspection brief)	M	B

ID	Feature	Judge-visible impact	Overlap with public repos	Effort	Tier
U-08	Reproducibility bundle (re-derivable decisions for appeals)	Medium–High	Partial (signed evidence pack)	S–M	B
U-09	Floor-plan area reconciliation (+ unit sanity check)	Medium	Partial (CrashCoders room detection)	M	B
U-10	Inter-officer consistency index	Medium	None seen	M	C
U-11	Governed calibration queue (feedback never edits rules silently)	Medium	Contrasts CrashCoders instant recalibration	S–M	C

Parity items (needed so you do not look weaker than Pramana): P-01 … P-08 in 4.3.

U-01 Layered rule packs — base handbook + addenda overlays

What. Represent the regulation the way AICTE actually publishes it: a base handbook plus dated addenda/corrigenda/instructions. `evaluate()` receives a **resolved rule stack for a cycle**; every threshold carries provenance ("value 20% came from `ADD_2025_26_POP`, replacing the base 15%").

Why it is unique. The only competitor README that mentions addenda (Pramana) describes a versioned ruleset built from the handbook and its addenda; none shows overlays with effective-cycle resolution, per-value provenance, or impact analysis (S3). The handbook *does* change mid-cycle **[V/R]** (S13, S14, S25), so this is a requirement, not decoration.

Design.

```
backend/rules/aph_2024_27/
  index.json                     # ordered list of layers + status (DRAFT|VERIFIED|PUBLISHED)
  base/2024-25/{checklists,rules,norms}.json
  overlays/
    addendum_2025_26_pop.json
    addendum_2026_27_supernumerary.json   # sample content, verified:false until checked
    instruction_<date>_<slug>.json        # AICTE "authorized instructions" (brief R08)
```

Overlay format:

```
{
  "overlay_id": "ADD_2026_27_SUPERNUMERARY",
  "kind": "ADDENDUM",
  "applies_from_cycle": "2026-27",
  "applies_to_cycle": null,
  "order": 20,
  "status": "DRAFT",
  "source_ref": {"doc": "Addendum/Corrigendum 2026-27 of APH 2024-27", "clause": "[?]", "page": null, "verified": false},
  "ops": [
    {"op": "set", "path": "/supernumerary/management_design_other/max_pct_of_sanctioned", "value": 25.0},
    {"op": "set", "path": "/eligibility/require_nil_deficiency_on_self_disclosure", "value": true}
  ]
}
```

Resolver:

```
import copy, json, hashlib

def in_cycle(ov, cycle):                      # "2025-26" strings sort lexicographically
    return ov["applies_from_cycle"] <= cycle and (ov["applies_to_cycle"] is None or cycle < ov["applies_to_cycle"])

def resolve_rules(base: dict, overlays: list[dict], cycle: str):
    resolved, provenance = copy.deepcopy(base), {}
    for ov in sorted((o for o in overlays if in_cycle(o, cycle)), key=lambda o: (o["applies_from_cycle"], o["order"])):
        for op in ov["ops"]:
            apply_op(resolved, op)            # set/add/remove/replace via JSON Pointer; schema-validated
            provenance.setdefault(op["path"], []).append(ov["overlay_id"])
    digest = hashlib.sha256(json.dumps(resolved, sort_keys=True, separators=(",", ":")).encode()).hexdigest()
    return resolved, provenance, digest
```

Conflict rule: two overlays writing the same pointer in the same cycle are rejected unless the later one declares `supersedes: <overlay_id>`.

UI — "Rule Lens".

- Hover any threshold on a finding → popover: *base value* → *overlay* → *resolved value*, clause anchor, `verified` chip.
- **Cycle slider** (2024-25 · 2025-26 · 2026-27) on an application: findings re-render; flips are highlighted. This is the money shot of the demo.

Acceptance tests.

1. Same facts + same cycle + same stack → identical `ruleset_sha256` and identical findings.
2. Removing an overlay restores base behaviour exactly.
3. Overlay with unknown pointer or wrong type fails schema validation.
4. Conflicting overlays rejected.
5. Golden test: PoP 11/60 = 18.3% → flagged in 2024-25 (15% cap), passes in 2025-26 (20% cap) **[R]**.

Honest limitation. You will not have the verified addendum text on day one. Ship the mechanism with *sample overlays marked* `verified:false` and say so; the contribution is the mechanism and the workflow. Replace sample values as you verify them.

Effort M · **Tier** A.

U-02 Impact simulator + maker-checker publish gate

What. Before any new layer goes live, run it against **all stored applications** (or the benchmark suite) and show exactly who flips.

Output (example layout).

Institution	Requirement	Old	New	Reason	Evidence
(synthetic)	PoP share, UG E&T	`Mismatch Detected`	`Verified`	cap 15% → 20%	PoP 11/60
…					

Summary tiles: `n_flipped`, `newly_clear`, `newly_deficient`, top affected programmes/states, runtime.

Mechanics.

```
def impact(candidate_overlay, population_facts, cycle, as_of):
    cur, _, _ = resolve_rules(BASE, OVERLAYS, cycle)
    new, _, _ = resolve_rules(BASE, OVERLAYS + [candidate_overlay], cycle)
    flips = []
    for inst_id, facts in population_facts.items():
        a, b = evaluate(facts, cur, cycle, as_of), evaluate(facts, new, cycle, as_of)
        flips += diff_findings(inst_id, a, b)          # keyed by requirement_id
    return summarize(flips)
```

Because `evaluate()` is a pure function over cached facts (your v2.0.0 design), 1,000 applications is a seconds-level job **[A — measure it]**.

Maker-checker.

- `Rules_Admin` A drafts → `Rules_Admin` B (different user) approves; self-approval blocked.
- Publication = new immutable layer + signature over `ruleset_sha256`; rollback = publish a superseding layer (never edit).
- Publish blocked if any changed clause has `verified:false`, unless an override with justification is recorded.
- Every step is an `audit_trail` row.

Acceptance tests. Deterministic flip list; self-approval returns 403; published layer cannot be mutated; rollback restores previous resolved hash.

Effort M · **Tier** A.

U-03 Counterfactual remediation engine

What. For every deficiency, compute **minimal quantified fixes**. Detection-only tools stop at the flag; officers drafting an LoD and institutions rectifying both need "how much, of what".

Worked example (use in the demo). UG Engineering, `S = 1080` (annual intake 270 × 4 yrs), `fsr = 20`, `R = 9`. Present: Professor cadre incl. Principal **4**, Associate **10**, Assistant **30** (total 44). Required (C-02): **6 / 12 / 36** (total 54).

Option	Action	Result	Caveat
(a) Hire	+2 Professors, +2 Associates, +6 Assistants	Meets 54 at intake 270/yr	Needs qualified, verifiable hires

Option	Action	Result	Caveat
(b) Reduce intake	Intake cap `S_max = min(4/1, 10/2, 30/6) × 9 × 20 = 720` → 180/yr	No hires needed	Large cut; cadre-limited (Professor cadre binds)
(c) Mixed	+2 Professors → `S_max = min(6, 5, 5) × 180 = 900` → 225/yr	One cadre fix + moderate cut	
(d) Relaxation check	If 3-yr avg admissions ≤ 50% of sanctioned → requirement ×0.75 = 40.5 → 41	Present 44 ≥ 41 → no deficiency	Only if admission history qualifies (C-05)

Assumption [A]: no substitution across cadres unless the clause allows it.

Algorithm (closed forms because requirements are monotone in intake).

```
def max_intake_supported(have, fsr, ratio=(1,2,6), relax=False):
    R = sum(ratio); unit_max = min(have[i] / ratio[i] for i in range(3))   # have = (prof_cadre, assoc, asst)
    return unit_max * R * fsr / (0.75 if relax else 1.0)                  # S_max

def remedies_fsr(prog, rules):
    need = ug_cadre(prog.S, rules.fsr, relax=prog.relax)
    gap = {k: max(need[k] - prog.have[k], 0) for k in ("professor_cadre_incl_principal", "associate", "assistant")}
    opts = []
    if any(gap.values()): opts.append({"type": "HIRE", "gap": gap})
    s_max = max_intake_supported((prog.have_prof, prog.have_assoc, prog.have_asst), rules.fsr, relax=prog.relax)
    if s_max < prog.S: opts.append({"type": "REDUCE_INTAKE", "S_max": s_max, "annual": s_max / prog.years})
    return opts
```

Other rule families:

- **PC / printer / plotter:** `+N PCs = ceil(required − have)`; printers `ceil(0.05 × PCs)` **[?]**.
- **Bandwidth:** upgrade to next tier **or** cap total intake at the tier boundary (e.g., ≤ 300 for 100 Mbps) **[?]**.
- **Area deficit:** `+X sq.m` or intake reduction to `Y`.
- **Expired/missing document:** list exact artefact to obtain + issuer type. Do **not** invent undertaking routes; only show one where the handbook text provides it **[?]**.
- **Combined:** search for the smallest intake reduction that clears *all* deficiencies (monotone → binary search per programme).

Output language. Every option is labelled *"advisory — computed from the loaded rule stack; final requirements are determined by AICTE".*

Acceptance tests. (1) Applying option (a)/(b)/(c) to the inventory and re-running `evaluate()` yields zero faculty deficiency (closure property). (2) Monotonicity: increasing intake never reduces requirements. (3) Remedy text cites the same `source_ref` as the finding.

Effort M · **Tier** A.

U-04 Intake Planner (applicant-side inverse of the evaluator)

What. An institution enters a *target* ("+60 CSE intake" or "new programme") and gets the full requirement set — faculty by cadre, PCs, printers, bandwidth tier, area, required documents, whether the application will route to EVC — versus current inventory, with a gap list.

Why it matters. The 2026-27 addendum reportedly demands "nil deficiency based on self-disclosure" **[R]** (S14). A planner reduces deficiencies *at source*, which is the point of "earliest stage" validation in the brief.

Design. `required_inventory(target, rules)` is the same code the evaluator uses, run backwards; `POST /api/planner/intake {inst_id, changes[]}` returns `{requirements, current, gaps, routing, documents_required, doc_expiry_warnings}`.

Property test (closure). For random targets: set inventory exactly equal to the planner's requirements → `evaluate()` returns **no** eligibility findings.

UI. Sliders for intake, programme add/remove; live table (green/amber/red); "Export plan" PDF.

Effort M · **Tier** B.

U-05 Claim coverage ("declared vs proven") + risk-based inspection focus

What. Decompose the application form into **atomic claims** (fire NOC valid till D, N regular faculty, M PCs, A acres, NBA valid, …). Each claim has an **evidence level**:

Level	Meaning
L0	Declared only

Level	Meaning
L1	Document uploaded, unsigned, internally consistent
L2	Document consistent across form ↔ PDF ↔ historical record
L3	Cryptographically anchored (DSC chain / issuer-signed QR / DigiLocker XML)
L4	Confirmed at source (T2) or on site (EVC)

Outputs. (1) **Claim Coverage %** = mandatory claims at ≥ L2 (and ≥ L3 where available); (2) **Inspection Focus List** — top-N claims ranked by `weight × (1 − evidence_strength) × anomaly_factor`, each with an explainable reason. Directly useful when deciding which cases/claims go to EVC.

Overlap. Partial with Pramana's proof-strength ladder (S3). Your distinct angle: the **coverage KPI** and **inspection sampling** tied to self-disclosure.

Acceptance tests. Coverage monotonic with added evidence; ranking stable under reordering; every ranked item shows the formula inputs.

Effort M · **Tier** B.

U-06 Verification tiers + feasibility matrix (+ DigiLocker consent-pull)

What. An honest model of *how* each document can be verified, shown in the product and in the deck. It prevents the failure mode seen in the field — mock registries labelled "live" (S7) — and respects the brief's rule not to assume access to restricted databases **[V]** (S1).

Tiers.

Tier	Method	Example	Prototype status
T0	Declared	Form field	real
T1	Cryptographic, offline	DSC chain to CCA root; issuer-signed QR; DigiLocker XML integrity HMAC	**real** (P-01)
T2	Authorised issuer interface	DigiLocker requester APIs (list issued docs, fetch file/XML) **[V]** (S21); authorised issuer APIs	**simulated, watermarked**
T3	Public portal lookup by officer	Council/registry lookup the officer performs; result recorded with source, time, screenshot hash	real workflow, manual
T4	Cross-document / historical consistency	Form ↔ PDF ↔ `institutions_historical`	real
T5	Physical	EVC visit	out of scope

Feasibility matrix (fill and keep updating; every "access basis" cell is [?] until confirmed with the issuer/AICTE).

Document	Issuer	Best tier today	Access basis [?]	Prototype	Fallback
Affidavit-2	AICTE portal (generated) + notary/e-stamp	T4 vs application record; T1 if DSC	Own record	real	Manual
Fire Safety Certificate	State fire department	T1 if signed/QR; else T3/T4	Varies by state	simulated	Manual confirm
Occupancy/Completion	Local authority	T4	Varies	simulated	Manual
Structural stability	Competent authority/engineer	T4	Varies	simulated	EVC
Land documents	State land records / registrar	T2 where DigiLocker-integrated, else T3/T4	Varies by state	simulated	Manual
Architect certificate	Council of Architecture	T3	Public registry lookup	simulated	Manual
Audited accounts	Chartered Accountant (UDIN)	T3	Public lookup	simulated	Manual
TER receipt	AICTE internal ledger	T2 (internal)	Own system	simulated	Manual
Accreditation (NBA/NAAC)	Accrediting body	T3	Public lists	simulated	Manual
Faculty degrees	University/NAD/DigiLocker	T2 with consent	Consent flow	simulated	Manual

Every verification result stores: `source`, `method`, `tier`, `queried_at`, `response_sha256`, `result`, `status`, `simulated: bool`, `attempts`. Simulated results render with a persistent "SIMULATED SOURCE" watermark.

Coverage panel. "For this application: 7 of 10 required documents are verifiable at T1/T2 *in production today* (3 simulated in this demo)."

Consent-pull flow (design). Applicant clicks *Fetch from DigiLocker* → consent → requester API returns issued document + integrity HMAC → document enters the packet pre-marked **T2**. In the prototype this is a simulated adapter labelled as such.

Effort S–M · **Tier** A.

U-07 Disclosure Auditor — public-website three-way check

What. AICTE mandates that institutions disclose, at a prominent place on their website: department-wise infrastructure with approved courses/intake; faculty details (qualification, tenure, experience) with Student:Faculty Ratio and Cadre Ratio; and audited financial statements for the last three years **[V]** (S12, cited as Annexure-18). The Auditor fetches the institution's **public** disclosure pages and compares them with the application.

Why it is unique. No competitor README shows a check against the public disclosures AICTE itself requires; v2.0.0 only receives `website_*` booleans from the form. This uses an open, legal data source — no restricted database needed (brief R17).

Pipeline.

1. **Fetch** (allow-listed domain from the application record; `robots.txt` respected; rate-limited; identifying user-agent; no login, no CAPTCHA bypass). Prototype default: **local synthetic sites**; live-fetch behind a flag.
2. **Extract:** presence of the three mandatory sections; faculty table (names, designation, qualification); SFR/cadre statements; infrastructure table; links to audited statements (FY labels); optional short-video presence.
3. **Evidence snapshot:** URL, retrieval timestamp, HTML SHA-256, optional screenshot hash.
4. **Compare:**
   - *Presence:* items (i)–(iii) present? → `disclosure_warnings[]` (advisory only).
   - *Three-way:* website roster ↔ application roster (set difference + fuzzy name match) → count of names missing on either side; website SFR ↔ computed SFR.
   - *Freshness:* "last updated" cues; audited statements cover the last 3 FYs.
5. **Output** to the **Disclosure tab**, never to eligibility findings (preserves your Annexure-18 separation).

Guardrails. Public pages only; store derived facts + snapshot hash (not mirrored content); advisory wording; officer sees why a name did not match (fuzzy score).

Acceptance tests. 20 synthetic sites with seeded gaps (missing section, stale year, roster mismatch, mismatching SFR) + compliant sites: report precision/recall **and false-warning rate** on compliant sites. A planted JS-rendered page must degrade to *Manual Review*, not to a warning.

Effort M–L · **Tier** B.

U-08 Reproducibility bundle (appeal-ready determinism)

What. A downloadable bundle that lets anyone re-derive a finding:

```
bundle_<inst>_<run>.zip
  manifest.json      # run_id, cycle, as_of_date, evaluator_version (git sha / image digest), hashes, signature
  facts.json         # normalised facts used (PII-minimised)
  rules_resolved.json + provenance.json
  findings.json
  reproduce.py       # re-runs evaluate() offline and compares findings_sha256
  README.md
```

Why. Decisions can be appealed (Standing Hearing/Appellate Committees are named in your spec). "Same facts + same resolved rules + same evaluator ⇒ same result" is something a statutory body can rely on. Partial overlap with Pramana's signed evidence pack (S3); your angle is **executable** reproducibility.

Acceptance tests. `reproduce.py` exits 0 on the shipped bundle; flipping one byte in `facts.json` or `rules_resolved.json` makes it exit non-zero and name the failing hash.

Effort S–M (after U-01 and C-26) · **Tier** B.

U-09 Floor-plan area reconciliation (+ unit sanity check)

What. The brief explicitly includes floor plans among infrastructure evidence **[V]** (S1); an earlier AICTE approval-portal statement also asked for "digital dimension tracking" **[V]** (S24). v2.0.0 has no floor-plan module.

Pipeline (indicative, never conclusive).

1. Classify plan sheet vs area-statement sheet.
2. Extract room labels and areas from the PDF text layer (or OCR): `Classroom 66.0 sq.m`, `Lab-3 72.5 m²`.
3. Parse the **carpet/built-up area statement** table if present.
4. Aggregate by category (instructional, administrative, amenities) — category definitions per the handbook appendix **[?]**.
5. Reconcile with the form's declared areas and `institutions_historical.approved_built_up_sqm`; compare minimum classroom/lab areas (`66 sq.m` in v2.0.0 **[?]**).
6. **Unit sanity check:** `1 sq.m = 10.7639 sq.ft`. If a plan's totals divided by declared totals ≈ 10.76 (±3%), flag "possible sq.ft/sq.m mix-up" — a common, explainable error.
7. Optional CV: detect scale bar + closed polygons to estimate area; always labelled *indicative*.

Acceptance tests. Synthetic plans (generated with area tables) with 3 seeded errors: unit mix-up, sub-minimum classroom, total below declared by > 10%. A plan with no parseable area statement → *Manual Review Required*.

Effort M · **Tier** B.

U-10 Inter-officer consistency index

What. The brief's expected outcomes include "more consistent application of configurable regulatory checklists" **[V]** (S1). Measure it.

- Maintain a **gold set** of ~20 seeded cases; periodically present them to officers (blinded).
- Compute agreement (Cohen's/Fleiss' κ) and per-rule **override rate by region/officer**.
- Dashboard highlights rules with high disagreement (usually ambiguous clauses or UI problems).

Ethics/scope. Aggregate and rule-centred, not a performance ranking of individuals; document that in the UI. In the prototype, ship the metric definitions and synthetic data only.

Effort M · **Tier** C.

U-11 Governed calibration queue

What. Officers mark a flag *Confirmed* or *False alarm*. Labels go to a **calibration queue**. A weekly job proposes threshold/weight changes, **back-tests** them on all labelled cases (shows effect on FAR and recall), and routes the proposal through U-02's maker-checker as an overlay.

Contrast. CrashCoders' README says officer feedback immediately recalibrates scores (S4); ungoverned feedback can drift or be gamed. Here feedback **never** edits rules or thresholds directly.

Effort S–M (reuses U-01/02) · **Tier** C.

U-12 Safety panel — False-Accusation Rate, abstention, confidence intervals

What. A first-class page (and slide) showing:

- **FAR** = share of *clean/benign* packets that receive an adverse status (`Mismatch Detected` or `Expired` where truly valid).
- **Review burden** = share of clean packets routed to `Manual Review Required` (workload cost of being careful).
- **Abstention/coverage curve:** accuracy vs share of cases auto-decided as the confidence threshold moves.
- **Per-signal precision/recall** (ELA, font, QR, pHash/ORB, revision diff), including *standalone* ELA.
- **Wilson 95% intervals** on every rate; "rule of three" upper bound when zero events (`≈ 3/n`; e.g., 0 false accusations in 35 clean packets ⇒ FAR upper bound ≈ 8.6%).
- **Worst-case list:** the clean packets that came closest to an accusation.

Why. Adverse regulatory action can harm institutions; a platform that measures its own harm is credible. Competitors report "0 false alarms" on their own generated sets (S3, S4) without intervals.

Effort S · **Tier** A.

4.3 Parity items (do these so you are not weaker than the strongest public repo)

ID	Item	How	Effort
P-01	Real DSC verification against CCA India roots	`pyHanko` with a trust store of official CCA root/intermediate certificates (verify hashes of the files you download); optional OCSP/CRL; **test on a real DSC-signed government PDF (e.g., e-Gazette)**; show "applicant's own DSC ≠ issuer authentication" distinction (Pramana does, S3)	M

ID	Item	How	Effort
P-02	PDF revision-level diff	`pikepdf` to enumerate incremental updates; rebuild each revision; diff text/objects inside the value bbox; message: "value changed in revision 2: '30-06-2025' → '30-06-2027'"	M
P-03	Encryption at rest, authenticated page access, parser sandbox	C-15 + C-18	M
P-04	Retention/purge + audit-chain anchoring	C-16 + C-17	S–M
P-05	Extraction stack	Text+layout classifier with confidence; `rapidocr-onnxruntime` + Tesseract (`eng+hin+tam`); **optional** local small-LLM fallback with schema-constrained output and *found-on-page* grounding (C-18); deterministic mode is default	M
P-06	Cross-application linkage	Shared PAN digest / certificate number / file hash / photo fingerprint → small graph view	S–M
P-07	Tamper Demo Mode	Clean vs tampered twin side-by-side (idea from CrashCoders, S4); one click	S
P-08	Applicant readiness check + rectification-scope check	Pre-submission run; on re-upload flag edits outside the requested deficiency scope	S

---

5. Specification v2.1 deltas

5.1 Architecture (changes marked ★)

```
                    STAGE 1  (once per upload; results cached)
 Upload ─► sandbox ★ ─► ingest_extract ─► doc_forensics ─► infra_vision ─► external_gateway ★(tiers)
            │              classifier ★      revision diff ★    floor_plan ★     simulated/real watermark ★
            │              rapidocr/tess     DSC vs CCA roots ★ pHash+ORB
            ▼                                                   disclosure_auditor ★ (public web)
      encrypted store ★  ──────────────────────────►  packet_facts (versioned, hashed)
                                                          │
                    STAGE 2  (pure function, ms)          ▼
 rule_layers ★ (base + overlays) ─► resolve(cycle) ★ ─► evaluate(facts, stack, cycle, as_of)
                                                          ├─► findings (completeness ⊥ verification ★)
                                                          ├─► remedy_solver ★  /  intake_planner ★
                                                          ├─► claim_coverage ★
                                                          └─► evaluation_run ★ (rules applied, records checked, hashes)
 Governance: draft ─► impact simulator ★ ─► maker-checker ★ ─► signed layer ─► audit chain + checkpoints ★
 Human gate: state machine ★ (system actor cannot issue adverse outcomes)
 Outputs: dashboard · officer workbench · applicant portal · reproducibility bundle ★ · safety panel ★
```

5.2 Repository tree additions

```
backend/
  security/{auth.py, crypto.py, sandbox.py, retention.py, audit_chain.py}      ★
  rules/aph_2024_27/{index.json, base/2024-25/, overlays/}                     ★
  modules/{rule_layers.py, impact.py, remedy_solver.py, intake_planner.py,
           claim_coverage.py, disclosure_auditor.py, floor_plan.py,
           verification_tiers.py, reproducibility.py, calibration.py}          ★
  modules/doc_forensics/{structure.py, fonts.py, ela.py, qr.py, dsc.py}       (split for tests)
  state_machine.py                                                            ★
  tests/{unit/, property/, golden/, redteam/, security/}                       ★
data_generator/
  generate_synthetic_packets.py   # --quick / --full / --template-shift / --redteam ★
  generate_synthetic_sites.py     # for U-07 ★
  benchmark.py                    # prints measured vs target, CIs, FAR ★
docs/{claims_register.md, verification_matrix.md, aph_verification_checklist.md}  ★
Dockerfile, docker-compose.yml    # one-command, offline-capable ★
```

5.3 Rule-pack layout and manifest

```
// index.json
{
  "handbook": "AICTE_APH_2024_2027",
```

```
  "layers": [
    {"id": "BASE_2024_25", "kind": "BASE", "path": "base/2024-25", "applies_from_cycle": "2024-25", "status": "VERIFIED"},
    {"id": "ADD_2025_26_POP", "kind": "ADDENDUM", "path": "overlays/addendum_2025_26_pop.json", "applies_from_cycle": "2025-26",
"status": "DRAFT"},
    {"id": "ADD_2026_27_SUPERNUMERARY", "kind": "ADDENDUM", "path": "overlays/addendum_2026_27_supernumerary.json",
"applies_from_cycle": "2026-27", "status": "DRAFT"}
  ],
  "default_cycle": "2026-27",
  "source_pdf_hashes": {"APH_2024_27.pdf": "<sha256 of the file you used>", "ADDENDUM_2025_26.pdf": "<sha256>", "ADDENDUM_2026_27.pdf":
"<sha256>"}
}
```

Remove the placeholder `source_pdf_sha256` in v2.0.0 (`e3b0c442…b855` is the SHA-256 of the **empty string**, so it identifies no real PDF — replace with the real file hash or leave `null`).

5.4 Schema additions

```
CREATE TABLE users (user_id INTEGER PRIMARY KEY, username TEXT UNIQUE NOT NULL, pw_hash TEXT NOT NULL,
  role TEXT NOT NULL, region TEXT, is_active INTEGER DEFAULT 1, totp_secret_enc BLOB);

CREATE TABLE documents (doc_id TEXT PRIMARY KEY, inst_id TEXT NOT NULL, slot TEXT NOT NULL, sha256 TEXT NOT NULL,
  enc_alg TEXT NOT NULL, wrapped_dek BLOB NOT NULL, nonce BLOB NOT NULL, key_id TEXT NOT NULL,
  uploaded_at TEXT NOT NULL, legal_hold INTEGER DEFAULT 0, purge_after DATE, purged_at TEXT);

CREATE TABLE rule_layers (layer_id TEXT PRIMARY KEY, kind TEXT NOT NULL, applies_from_cycle TEXT NOT NULL,
  applies_to_cycle TEXT, order_no INTEGER NOT NULL, status TEXT NOT NULL,   -- DRAFT|APPROVED|PUBLISHED
  payload_json TEXT NOT NULL, payload_sha256 TEXT NOT NULL, maker TEXT, checker TEXT, signature TEXT, created_at TEXT);

CREATE TABLE verification_results (vr_id INTEGER PRIMARY KEY, inst_id TEXT NOT NULL, requirement_id TEXT NOT NULL,
  source TEXT NOT NULL, method TEXT NOT NULL, tier TEXT NOT NULL, queried_at TEXT NOT NULL, response_sha256 TEXT,
  result_json TEXT, status TEXT NOT NULL, simulated INTEGER NOT NULL, attempts INTEGER DEFAULT 1);

CREATE TABLE claims (claim_id TEXT PRIMARY KEY, inst_id TEXT NOT NULL, field TEXT NOT NULL, value TEXT NOT NULL,
  weight REAL NOT NULL, evidence_level INTEGER NOT NULL, evidence_refs_json TEXT);

CREATE TABLE disclosure_audits (audit_id INTEGER PRIMARY KEY, inst_id TEXT NOT NULL, url TEXT NOT NULL,
  fetched_at TEXT NOT NULL, html_sha256 TEXT NOT NULL, findings_json TEXT NOT NULL);

CREATE TABLE chain_checkpoints (log_id INTEGER PRIMARY KEY, head_hash TEXT NOT NULL, mac TEXT NOT NULL, created_at TEXT NOT NULL);

CREATE TABLE calibration_proposals (proposal_id INTEGER PRIMARY KEY, created_at TEXT, change_json TEXT NOT NULL,
  backtest_json TEXT NOT NULL, status TEXT NOT NULL, layer_id TEXT);
```

Plus `evaluation_runs` (C-26) and the history tables (C-27). Add `source_ref_verified` column to any table caching rule metadata.

5.5 API changes and additions

#	Method & route	Roles	Notes
1'	`POST /api/auth/login {username,password,totp?}`	Public	Role from DB (C-13)
4'	`GET /api/applications/{id}/packet`	All	Adds `claims` , `coverage_pct` , `verification_results` (with tier/simulated); PAN shown as last-4 only
6'	`POST /api/applications/{id}/evaluate {cycle, as_of_date, what_if_overrides?}`	All	Returns `completeness` ⊥ `verification_status` , `rules_applied` , `ruleset_sha256`
8'	`POST /api/applications/{id}/scrutiny-action`	Officers	State-machine enforced (C-29)
11'	`POST /api/rules/publish`	`Rules_Admin`	Only for an **approved** draft (maker ≠ checker)
13	`GET /api/rules/resolve?cycle=2026-27`	All	Resolved rules + provenance + hash
14	`POST /api/rules/drafts` · `POST /api/rules/drafts/{id}/approve`	`Rules_Admin`	Maker-checker
15	`POST /api/rules/impact {draft_id, cycle}`	`Rules_Admin` , officers (read-only)	Flip table
16	`POST /api/applications/{id}/remedies`	Officers, Applicant (own)	Remedy options (U-03)

#	Method & route	Roles	Notes
17	`POST /api/planner/intake`	Applicant	Intake Planner (U-04)
18	`POST /api/applications/{id}/disclosure-audit` · `GET …/disclosure-audit`	Officers	U-07
19	`GET /api/applications/{id}/reproducibility-bundle` · `POST /api/reproduce`	Officers	U-08
20	`GET /api/verification-matrix`	All	U-06 matrix + live coverage
21	`GET /api/safety/metrics`	Officers, `Rules_Admin`	U-12
22	`POST /api/calibration/proposals` · `…/{id}/approve`	`Rules_Admin`	U-11
23	`GET /api/pages/{doc_id}/{n}`	Authorised	Replaces public `/static` page images (C-15)

5.6 UI additions

- **Rule Lens** popover + **cycle slider** (U-01).
- **Impact Simulator** screen with flip table and maker-checker panel (U-02).
- **Remedy panel** under each deficiency; **Intake Planner** for applicants (U-03/04).
- **Verification Tier badge** (T0–T5) and **SIMULATED SOURCE** watermark; **Verification Coverage** card (U-06).
- **Disclosure tab** (U-07) separate from eligibility.
- **Claims & Coverage** view with **Inspection Focus list** (U-05).
- **Safety panel** (U-12), **Reproduce** button (U-08).
- **Tamper Demo Mode** toggle (P-07); EN/HI language toggle (cheap).
- Keep a single, calm officer path: *Queue → Application → Findings → Evidence → Action*. Everything above is reachable from there; do not add more top-level tabs.

5.7 `requirements.txt` v2.1 (indicative; pin versions you test)

```
fastapi, uvicorn[standard], pydantic>=2, argon2-cffi, pyjwt, cryptography, python-multipart
pypdfium2, pdfplumber, pikepdf, pypdf, pyhanko, pyhanko-certvalidator
pillow, opencv-python-headless, zxing-cpp, imagehash, numpy
rapidocr-onnxruntime, pytesseract, onnxruntime
rapidfuzz, python-dateutil, jsonpointer, jsonschema
httpx, selectolax
reportlab, qrcode
pytest, hypothesis
```

(System packages in the Docker image: `tesseract-ocr tesseract-ocr-hin tesseract-ocr-tam ffmpeg`.)

---

6. Corrected evaluation protocol — how to show *measured* results honestly

6.1 Principles

1. **Targets are not results.** Show `Target | Measured | 95% CI | n` side by side.
2. **"Held-out" must mean something.** Different generator seeds **and** layout families the extractor never saw during development.
3. **Hard negatives are first-class.** A system that never accuses a genuine applicant is the credible one.
4. **Intervals everywhere.** With 15 positives and 35 negatives, point estimates alone mislead.
5. **No tuning on the test set.** Freeze code/thresholds at a tagged commit before the test run.
6. **Publish the generator and seeds** so anyone can rerun (`make eval SEED=…`).
7. **Say plainly that data is synthetic.** The brief requires authorized/anonymized/synthetic data **[V]** (S1); claim generalisation only as far as you tested it.

6.2 Dataset design

Split	Content	Purpose
Dev (15+ packets)	Layout families L1–L3, seeds 1–100	Calibrate thresholds (pHash/ORB, ELA, font)

Split	Content	Purpose
Test (50+ packets; increase to 300+ since generation is cheap)	L1–L3 with **new seeds + L4–L5 unseen families** (different fonts, margins, header wording); ≥ 20% scanned branch with skew/noise/JPEG q 40–80	Headline numbers
Hard negatives (≥ 10, inside test)	CamScanner/iLovePDF scans; missing EXIF; legitimate relaxation (≤ 50% admissions); valid structural cert on a 33-year-old building; **exact 30-year and exact 5-year anniversaries (C-21)**; Devanagari font fallback; QR with a different but legitimate URL structure; same-Trust shared stock photo	Measure false accusations
Red-team (Section 6.5)	One packet per attack class	Coverage matrix, not a score
Multilingual	≥ 10 pages Hindi/Tamil labelled documents	OCR accuracy by script
Real-document sanity	**Government-published, openly available** DSC-signed PDFs (e.g., e-Gazette) for signature verification; *do not commit real institution certificates*. Use public EoA letters only locally to test field patterns	Shows the pipeline does not only work on your generator

6.3 The eight metrics (the brief's list, [V] S1)

#	Metric	Unit	Definition	CI method
1	Document-extraction accuracy	field	Exact match after normalisation; report digital / scanned / Hindi / Tamil separately; micro-average + per-field table	Cluster bootstrap by packet
2	Validation accuracy	requirement + packet	Agreement with ground truth per requirement **and** packet exact-match	Wilson (requirement), bootstrap (packet)
3	Anomaly-detection performance	packet and document	Precision / recall / F1; **per-signal** results (ELA, font, QR, revision diff, pHash/ORB); **FAR** and **review burden** on clean packets	Wilson
4	Successful source-verification rate	adapter × mode	**Status-mapping accuracy under fault injection:** 6 adapters × 6 modes (`normal, mismatch, expired, timeout, server_down, inconclusive`) → expected status matrix (36 cells); plus unhandled-exception count (must be 0) and p95 latency	Exact count
5	Discrepancy-identification reliability	injected discrepancy	Detected **and** localised: bbox IoU ≥ 0.5 with ground-truth region and correct evidence cited	Wilson
6	Average verification time	seconds/packet	Stage 1 and Stage 2 separately; median, p95; hardware stated; cold vs warm; pages/packet	Report n, IQR
7a	Alert timeliness	ms	Evaluation-complete → alert visible; p95; plus alert correctness (no alerts on clean packets)	Report n
7b	Reduction in manual effort	%	Measured time-and-motion (6.6), with accuracy recorded alongside	Paired test

6.4 What the numbers in v2.0.0 would look like with honest intervals

Computed with Wilson 95% intervals (z = 1.96):

v2.0.0 "target"	Counts	Point estimate	95% CI
Recall 100%	15/15	100%	**79.6% – 100%**
Precision 93.8%	15/16	93.8%	**71.7% – 98.9%**
Validation accuracy 98.0%	49/50	98.0%	**89.5% – 99.7%**
Discrepancy reliability 97.8%	44/45	97.8%	**88.4% – 99.6%**
FAR with 0 false accusations	0/35	0%	upper bound **9.9%** (Wilson); ≈ 8.6% (rule of three, `3/n`)

What to say. "No misses observed on 15 sabotaged packets (95% CI 79.6–100%); with the test set generator we can scale to 300+ packets to tighten this." That is stronger than an unqualified "100%".

6.5 Red-team coverage matrix (publish it, including blind spots)

ID	Attack	Primary detector	Expected status	Honest blind spot
A1	Later-revision text overlay on a date	Revision diff (P-02)	Manual Review (Medium) → **Mismatch** if corroborated by QR/source	Edit done before first save
A2	White box / cover rectangle over a value	Structure + text-layer vs render disagreement	Manual Review	Flattened image PDFs
A3	Font swap on a value	Font check (C-24)	Manual Review	Same font available
A4	Raster splice on scanned doc	ELA (supporting only)	Manual Review	Clean re-scan
A5	**QR regenerated with a self-consistent payload**	Needs issuer signature or source check	Verification Pending / T2 check	**Undetectable without issuer anchor**
A6	Whole PDF regenerated from Word	No issuer authentication present	Verification Pending (no T1/T2)	Content-level forgery with plausible data
A7	Print → scan laundering	Hygiene (Low) + missing issuer signal	Manual Review	Alterations before printing
A8	Photo of a printout	Same as A7	Manual Review	Same
A9	Lab photo reused, cropped, banner added (cross-Trust)	pHash + ORB/RANSAC	**Mismatch** (High)	Heavy re-shoot of same room
A10	EXIF stripped / spoofed GPS	EXIF check	Manual Review (stripped); spoofed = blind	Spoof within geofence
A11	DSC stripped from issuer document	Expected-signature absence	Manual Review / Verification Pending	Documents that never had a DSC
A12	Certificate copied from another institution	Cross-application linkage	**Mismatch** (High)	First-time reuse across years not in DB
A13	Date re-typeset with no trace	Cross-document + historical + source	Manual Review / Mismatch if a record contradicts	Consistent forgery across all docs
A14	Survey number differs form vs deed vs plan	3-way reconciliation	**Mismatch**	Same error replicated everywhere
A15	Affidavit says "uploaded" but document missing	Affidavit-vs-packet check	Completeness: MISSING	—
A16	Planted instruction text in document (if LLM used)	Grounding check	Ignored (0 followed)	Novel injection styles

State openly that **A5/A6 require issuer-side anchoring** (T1/T2). That is exactly why U-06 exists.

6.6 Time-and-motion protocol for metric 7b

1. Recruit volunteers; **disclose that they are not AICTE officers** unless you can recruit real ones (external-validity limit).
2. Same 10 packets per participant; counter-balanced order (manual-first vs assisted-first).
3. Manual = checklist + PDF viewer; assisted = platform.
4. Record **time and correctness** (a fast wrong decision is not a saving).
5. Report median and IQR of time per packet; paired Wilcoxon signed-rank; present as "estimated reduction in this study" with n.
6. Keep the old "18 min vs 2.5 min" as an *assumption* label only if you cannot run the study.

6.7 Timing methodology (metric 6)

`time.perf_counter()` around Stage 1 and Stage 2 separately; 3 warm-up runs; n ≥ 50 packets; stated CPU/RAM; no GPU; report median/p95; run `evaluate()` 1,000× on cached facts for the "ms" claim; separate digital-native from OCR-heavy packets.

6.8 Ablations (shows where value comes from)

Rules only → + cross-document reconciliation → + forensics → + external verification. Report FAR and recall at each step. This answers "do you need the heavy CV/forensics?" with data.

6.9 Reporting template (paste into `benchmark_report.md`)

Metric	Target	Measured	95% CI	n	Notes
1 Extraction accuracy					digital / scanned / HI / TA
2 Validation accuracy					
3 Anomaly P / R / F1					per-signal table below
3 FAR / Review burden					clean + hard negatives
4 Source-verification status mapping				36 cells	0 unhandled exceptions
5 Discrepancy reliability					IoU ≥ 0.5
6 Time (Stage 1 / Stage 2)					hardware
7a Alert timeliness					
7b Effort reduction					time-and-motion study

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7. Showing uniqueness

7.1 Positioning statements

15 seconds. "Other tools flag problems. We version the rulebook itself, show who is affected before a rule goes live, and tell officers and institutions exactly what would fix each deficiency — with verification tiers that never pretend to access databases we don't have."

60 seconds.

1. *Problem:* AICTE's portal checks self-disclosed data; document authenticity, cross-institution reuse and the public disclosures institutions must publish are checked manually. The handbook also changes every year through addenda.
2. *Solution:* A decision-support platform that validates applications against a layered, cited rule stack; authenticates documents with calibrated, corroborated signals; verifies at the issuer where legally possible and says clearly where not; and keeps every decision with the human officer.
3. *What's different:* layered rule packs with impact simulation; remediation options; honest verification tiers; a public-disclosure audit; measured false-accusation rate.
4. *Proof:* live demo with a tampered twin, a benign scan that is *not* accused, a cycle slider that flips a finding, and a reproducibility bundle that re-derives the decision.

7.2 Uniqueness proof table (claim → evidence → where to see it)

Claim	Evidence artefact	Demo moment
The rulebook changes mid-cycle and we model it	Overlay files + provenance map + `ruleset_sha256`	Cycle slider flips PoP finding (U-01)
We show the blast radius of a rule change	Impact flip table; maker-checker audit rows	Draft overlay → 3 flips → second admin approves (U-02)
We tell you how to fix it	Remedy options; closure test in CI	Apply option (c) → re-evaluate → clear (U-03)
We never claim access we don't have	Verification matrix + SIMULATED watermark	Tier badges, coverage card (U-06)
We don't falsely accuse	Safety panel with CIs; hard-negative set	Benign CamScanner PDF → Manual Review, not Mismatch (U-12)
We check what institutions must publish	Disclosure audit snapshot + comparison	Roster mismatch with website (U-07)
Decisions are reproducible for appeals	Bundle + `reproduce.py`	Flip one byte → failure (U-08)
Security isn't an afterthought	Encrypted blobs, server-side roles, sandbox, retention	Show 403 on role forgery; purge tombstone

7.3 Idea-deck plan (adapt to the official SIH template — I have not seen it)

#	Slide	Content
1	Problem and evidence	Self-disclosure-based checks; ghost faculty (Anna University reportedly found ~2,000 faculty positions wrongly filled in 2024-25 **[R]**, S20); handbook changes via addenda **[V/R]**

#	Slide	Content
2	Solution at a glance	3-sentence description + safeguards (advisory only, human gate, no restricted DB)
3	Architecture	Two-stage (extract once → pure-function evaluation) + governance layer (U-01/02)
4	What is different	U-01, U-02, U-03 with one visual each
5	Trust model	Verification tiers + calibrated, corroborated forensics + SIMULATED watermark
6	Evaluation plan	The eight metrics, FAR, CIs, red-team matrix
7	Feasibility, integration, security	Adapter contract, SSO placeholder, encryption, retention, licences
8	Impact and roadmap	Officer workload, institution self-service, rollout path

7.4 Five-minute demo script

Time	Beat	What the judge sees
0:00	Safeguard banner; one-line problem	"Decision support — not decision maker."
0:30	**Tamper Demo Mode** (P-07): clean vs tampered twin	Fire NOC visible expiry ≠ QR payload → Medium + High corroboration → `Mismatch Detected` with bounding boxes
1:15	**Benign scan**	CamScanner PDF, no EXIF → *Low chip + Manual Review*, **not** accused; point to Safety panel FAR
1:45	Cadre deficiency with **Rule Lens**	Hover threshold → base vs overlay vs resolved, clause anchor, `verified` chip
2:15	**Remedies**	Three options; apply (c); re-evaluate → deficiency cleared
3:00	**Cycle slider**	PoP 11/60 flips 2024-25 ↔ 2025-26
3:30	**Impact simulator + maker-checker**	Draft overlay → flip table → second admin approves; self-approve attempt blocked
4:15	**Verification matrix** and **Disclosure Auditor**	Tier badges; "3 faculty names on website not in application"
4:45	**Reproducibility** + Safety panel	`reproduce.py` passes; flip a byte → fails; CIs on screen

Fallbacks. Pre-recorded video of the whole flow; seeded DB; `docker compose up` works offline; screenshots of each beat.

7.5 Likely jury questions and answers

#	Question	Answer (short)
1	"Similar tools already exist."	Yes — several public prototypes target this PS. Ours differs in governing the rulebook (layered addenda, impact simulation, maker-checker), quantified remediation, honest verification tiers and a measured false-accusation rate. We adopted the parity items that matter (DSC verification, revision diff, encryption).
2	"Where is the AI?"	OCR/classification, near-duplicate image matching, optional grounded local LLM extraction, calibrated anomaly scoring. Final compliance decisions are deterministic and explainable by design (R19/R20).
3	"Does it call real government registries?"	Not in the prototype — adapters are simulated and watermarked. The matrix shows the production path per document (offline-verifiable signals, DigiLocker consent-pull, public lookups). We do not assume restricted databases (brief R17).
4	"What when the handbook changes?"	Publish an overlay; the impact simulator shows who flips; a second approver signs it; old decisions remain reproducible with the old stack.
5	"How do you avoid accusing genuine applicants?"	Corroboration rule, calibrated severity, benign-explanation display, abstention to manual review, and we publish FAR with intervals.
6	"Is data safe?"	Server-side roles, envelope encryption, sandboxed parsing, retention with audited purge, no raw government IDs, hash-chained audit with external anchoring.
7	"How does it integrate with the AICTE portal?"	Stateless evaluator + REST API + adapter contract; SSO placeholder; reads institutional records through authorised interfaces; PostgreSQL path.
8	"Your data is synthetic."	By the brief's own rules; we use template-shift splits, hard negatives, a red-team matrix with blind spots, plus real DSC-signed government PDFs for signature checks. Intervals reflect sample size.

#	Question	Answer (short)
9	"Why not just use an LLM?"	Regulatory outcomes need determinism, citations and reproducibility; LLMs are optional for field extraction only, with found-on-page grounding.
10	"Does it scale?"	Evaluation is pure and cacheable; Stage 1 runs once per upload in a queue; SQLite → PostgreSQL; measured throughput reported.
11	"Can institutions game the planner?"	The planner uses the same rules and doesn't bypass evidence checks; claim coverage and verification tiers still apply.
12	"How do you know your rules are right?"	Every rule carries a clause anchor and a `verified` flag; unverified rules route to Manual Review, never to deficiencies; changes go through maker-checker.
13	"What about regional languages?"	Hindi/Tamil OCR with confidence gating; low confidence routes to officers.
14	"What does 'reduced manual effort' rest on?"	A time-and-motion study with stated limits; until run, we label it an assumption.
15	"What can't you detect?"	Self-consistent forgeries without an issuer anchor (A5/A6); we say so and route to Verification Pending.
16	"Why separate website disclosure issues from eligibility?"	They are different regulatory consequences; advisory warnings must not drive a deficiency letter.

7.6 Claims register (keep in `docs/claims_register.md`; review before every presentation)

Topic	Allowed phrasing	Not allowed
Registry access	"Simulated adapters; production path via DigiLocker/authorised APIs"	"Live government API integration"
Accuracy	"Measured on N synthetic packets: X% (95% CI a–b)"	"100% accurate", "0 false alarms" without n and CI
Time saved	"Estimated Y% in a time-and-motion study with n volunteers"	A figure with no method
Blockchain	"Hash-chained, tamper-evident log with external checkpoints"	"Immutable consortium blockchain" unless it is one
Rules	"Cited; `verified` where checked against the PDF"	Page numbers you haven't checked
Fraud	"Preliminary indicator requiring human verification"	"Forgery detected"
ML models	Name the actual model and licence	Invented or fine-tuned model names
Regulation changes	"Rule stack supports addenda; sample overlays marked unverified"	Presenting unverified addendum content as AICTE text

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8. Build plan, priorities, cut list, risks

Effort units are **[A] rough estimates** for a small team (S ≤ 0.5 day, M ≈ 1–2 days, L ≥ 3 days, per person). Adjust to your team size. The finale date is not known to me; the plan is expressed in phases.

8.1 Phase 0 — before idea submission (reported deadline 28 Oct 2026; confirm)

Task	Done when
Replace unverified numbers/page refs with `[?]` / `null` (Section 9.2)	No unchecked citation in the deck
Fix the six credibility items: C-01, C-02, C-21, C-13, C-14, C-28	Each fix written into the deck's spec slide
Add U-01, U-02, U-03, U-06, U-12 to the *idea* (not the code)	Slides 3–6 of 7.3
Draft the Claims Register (7.6)	One page, reviewed by the whole team

8.2 Phase 1 — core platform (≈ 7 working days)

ID	Task	Effort	Depends
B-01	Schema v2.1 (users, documents, rule_layers, evaluation_runs, history tables), audit chain + checkpoints	M	—

ID	Task	Effort	Depends
B-02	Auth: argon2id, server-side roles, JWT, RBAC dependencies, object-level checks	M	B-01
B-03	Encrypted document store + authenticated page endpoint + sandboxed parsing	M–L	B-01
B-04	Rule layers: `index.json`, base + 2 sample overlays, resolver, schema validation	M	B-01
B-05	`evaluate(facts, stack, cycle, as_of)` with C-02, C-05, C-09, C-21, C-22	L	B-04
B-06	Ingest: classifier, digital/scanned branch, field extraction + bbox	L	B-03
B-07	Forensics: QR vs text, font check, ELA (supporting), revision diff (P-02)	L	B-06
B-08	Verification adapters with tiers, fault modes, SIMULATED watermark (U-06)	M	B-01
B-09	State machine for officer actions (C-29)	M	B-02
B-10	Synthetic generator with template shift + hard negatives + red-team packets	L	—
B-11	Benchmark script with CIs, FAR, per-signal metrics	M	B-05, B-07, B-10

8.3 Phase 2 — differentiators (≈ 7 working days)

ID	Task	Effort	Tier
D-01	Impact simulator + maker-checker + signed publish (U-02)	M	A
D-02	Remedy solver + closure tests (U-03)	M	A
D-03	Safety panel (U-12)	S	A
D-04	Rule Lens + cycle slider UI (U-01)	M	A
D-05	Intake Planner (U-04)	M	B
D-06	Disclosure Auditor on synthetic sites (U-07)	M–L	B
D-07	Claim coverage + inspection focus (U-05)	M	B
D-08	Reproducibility bundle (U-08)	S–M	B
D-09	Floor-plan reconciliation (U-09)	M	B
D-10	DSC verification against CCA roots on a real signed PDF (P-01)	M	parity

8.4 Phase 3 — hardening, evaluation, demo (≈ 5 working days)

Run the benchmark at a frozen commit; write `benchmark_report.md` (6.9); time-and-motion study (6.6) if possible; security tests (role forgery, path traversal, oversized upload, bomb image, injection text); one-command Docker run on a clean machine; rehearse the 5-minute demo (7.4) twice, record a backup video.

8.5 Cut list (drop in this order if time runs out)

1. U-10 consistency index (describe only).
2. U-11 calibration queue (describe only).
3. U-09 floor-plan CV (keep text-table parsing and unit sanity check only).
4. YOLO workstation counting (advisory; AGPL issue) — keep pHash/ORB and EXIF only.
5. U-07 live fetch (keep synthetic-site mode).
6. EVC video-duration check (cheap, but clause **[?]**).
7. SLA/undertaking tracker (overlaps PS 26262).

Never cut: server-side roles, encrypted storage, corroboration rule, completeness ⊥ verification split, cycle-aware rule resolution, FAR reporting, human-gate state machine, claims register.

8.6 Risk register

Risk	Likelihood	Impact	Mitigation
A cited norm/page is wrong in front of a judge	Medium	High	`verified` flags; unverified → Manual Review; checklist 9.2 completed

Risk	Likelihood	Impact	Mitigation
Pramana-level forensics depth expected	High	Medium	Parity items P-01/P-02; lead with governance + remediation + honesty
Over-scope; nothing works end-to-end	High	High	Phases and cut list; demo script drives priority
OCR/regex brittle on unseen layouts	Medium	Medium	Template-shift tests; confidence gating; optional grounded LLM
False accusations on benign scans	Medium	High	Corroboration rule; hard negatives; FAR reporting
Legal doubt about using public websites (U-07)	Low	Medium	Public pages only, robots.txt, advisory output, synthetic mode default
Licence challenge (AGPL)	Medium	Medium	C-19 replacements; licence table 9.3
Demo fails live	Medium	High	Offline Docker, seeded data, backup video
Simulated sources mistaken for real	Medium	High	Persistent watermark; matrix; claims register
Aadhaar-derived data handling concerns	Low	High	Do not persist Aadhaar-derived values; legal review [?]

8.7 Definition of done (acceptance checklist)

- [ ] `docker compose up` runs the full demo offline on a clean machine.
- [ ] Role forgery attempt returns 403; Applicant cannot see another institution.
- [ ] Page images require authentication.
- [ ] `evaluate()` is pure; 1,000 runs timed and reported.
- [ ] Golden tests for C-01, C-02, C-21 pass; property tests pass.
- [ ] Closure tests for remedies and planner pass.
- [ ] Maker ≠ checker enforced; published layers immutable.
- [ ] `benchmark_report.md` shows target vs measured with CIs, FAR, per-signal metrics.
- [ ] Red-team matrix published with blind spots.
- [ ] Reproducibility bundle verifies; tampering fails.
- [ ] Claims Register reviewed; every `[?]` either resolved or visibly flagged.

---

9. Appendices

9.1 Sources and reliability

ID	Source	URL	Used for	Reliability
S1	Official PS 26261 brief (mirror of sih.gov.in listing)	https://zaidsayyed.in/tools/sih-problem-statements/sih26261	Requirements, metrics, safeguards, six statuses	High for brief text (mirror); confirm on sih.gov.in
S2	SIH 2026 problem-statement index (mirror)	https://zaidsayyed.in/tools/sih-problem-statements	Idea counts, deadline, PS 26262 scope	Medium (unofficial)
S3	Pramana README	https://github.com/Anexus5919/pramana	Competitor features and reported results	README-level
S4	CrashCoders README	https://github.com/viraj-1709/aicte	Competitor features	README-level
S5	SmartValidate README	https://github.com/sathvik210508/AI-Powered-Application-Validation-Document-Authentication-Regulatory-Compliance-Platform	Competitor features, scope table	README-level
S6	SIH11 repo	https://github.com/sohamramshette/SIH11	Competitor features	README-level

ID	Source	URL	Used for	Reliability
S7	CipherCore Verify demo	https://nonutsayan.github.io/CipherCore-verify/	Competitor claims	Demo page
S8	VTU minimum-norms regulations reproducing AICTE faculty norms	https://www.ksit.ac.in/img/staff-list-2024-2025.pdf	FSR, cadre formulas, relaxation, 1:3:10	Medium-High (faithful copy)
S9	APH Appendix-7 copy	https://naac.skit.ac.in/AQAR-2023-24/Extended_Profile/3.2/AICTE_APH_Appendix-7.pdf	FSR tables	Medium-High
S10	APH 2018-19 Appendix-7 copy	https://diet.ac.in/wp-content/uploads/extended%20profiles/APH-18-19.pdf	Diploma formula (older edition)	Medium (older handbook)
S11	AICTE portal document lists (Chapter-1 categories)	https://aicte.gov.in/sites/default/files/New%20Institutes%20&%20Existing%20Institutes%20(Chapter-1%20Categories).pdf	Document list incl. 30-year stability certificate	High (aicte.gov.in)
S12	AICTE EoA 2026-27 letter (SPJIMR copy)	https://www.spjimr.org/wp-content/uploads/2026/06/aicte-approval-letter-2026-2027.pdf	Annexure-18 at page 180; mandatory disclosures	High (AICTE-generated)
S13	News on 2025-26 addendum (adjunct removed, PoP 20%)	https://edunovations.com/aicte-allows-20-faculty-at-technical-institutes-to-be-persons-from-industry/amp/ ; https://news.careers360.com/professor-of-practice-ugc-15000-university-80-iit-college-vacancy-appointments-jobs-aicte-guidelines-education-ministry	PoP/adjunct change	Medium (secondary)
S14	News on 2026-27 addendum	https://news.careers360.com/aicte-approval-process-handbook-2026-27-update-engineering-college-business-school-management-foreign-student-seats-minor-diploma/amp	Supernumerary 25%, nil-deficiency self-disclosure	Medium (secondary)
S15	APH 2024-27 adjunct/PoP text (VESIT copy)	https://cdn.vesit.ves.ac.in/assets/files/126663/aicte_ugc_guidelines_for_engaging_of_professor_of_practice.pdf	Original 15% wording	Medium (partial text)
S16	AICTE Affidavit sample	https://old.gcoej.ac.in/download/AICTE%20Affidavit%201-46226522477%20EOA%202025-260.pdf	Portal-generated affidavit structure	Medium-High
S17	APH 2024-27 launch coverage	https://news.careers360.com/aicte-launches-new-approval-handbook-removes-cap-intake-top-engineering-colleges-nirf-bba-bca-guidelines/amp	Three-year handbook; BBA/BCA/BMS scope	Medium
S18	Lok Sabha reply (e-governance since 2010)	https://eparlib.nic.in/bitstream/123456789/587954/1/91870.pdf	AICTE e-governance history	High
S19	AICTE EoA letter (self-disclosure deficiency)	https://igdtuw.ac.in/NAAC/Single-IMS/AICTE_Approval.PDF	Self-disclosure-based reports	High

ID	Source	URL	Used for	Reliability
S20	Aadhaar authentication for AICTE institutions	https://www.tribuneindia.com/news/india/aadhaar-use-okayed-in-tech-institutions/amp	Ghost faculty context; voluntary use	Medium
S21	DigiLocker requester API spec; explainer	https://cf-media.api-setu.in/resources/Requester-APISpecification-V1_12.pdf ; https://didit.me/blog/digilocker-api/	Consent-pull path	High (spec) / Medium (explainer)
S22	DU plans AI verification tool	https://theprint.in/india/du-plans-to-develop-ai-based-verification-tool-after-detecting-manipulated-documents-in-admissions/2991453/	Demand signal	Medium
S23	Generic AI document-verification vendor example	https://www.eklavvya.com/assessment-video-library/automate-document-verification-process/	Non-AICTE alternatives	Low-Medium
S24	Earlier AICTE approval-portal PS on sih.gov.in	https://www.sih.gov.in/sih2019ProblemStatementstudentInnovation	Dimension tracking, evaluator matching	High (portal), year unclear
S25	AICTE Approval Process 2025-26 page (links addendum)	https://www.aicte-india.org/bureaus/approval/approval-process-2025-26	Addendum existence	High
S26	APH 2024-27 text via document host	https://www.studocu.com/in/document/sant-gadge-baba-amravati-university/software-engineering/approval-process-handbook-2024-25-to-2026-27/80825336	Affidavit-2 sworn-before text	Low-Medium (third-party host)
S27	Public notice for Approval Process 2026-27	https://dteapengg.in/Documents/NOTIFICATIONS/AICTE%20Approval%20Process%202026-27_Fair%20Copy%20and%20Public%20notice%20for%20approval%20process%202026-27%20Merged_20251201_044937.pdf	Portal timeline; "amendments issued from time to time"	High

9.2 APH verification checklist (do this before the deck is final)

Open the **APH 2024-27 PDF**, the **2025-26 addendum**, and the **2026-27 addendum**. For each item record: clause, exact wording anchor, page, PDF SHA-256, your initials. Until done, keep the item `verified: false`.

#	Item	Where in your spec	Status
V-01	Annexure-18 clause list and page (letter cites p. 180)	`WEBSITE_DISCLOSURES_ANNEX_18`	☐
V-02	Annexure-14 competent fire authorities (all states)	`FIRE_AND_OCCUPANCY_ELIGIBILITY`	☐
V-03	Annexure-1 / Annexure-2 document lists, clause numbers, pages	`CATEGORY_CHECKLIST_SCOPE`	☐
V-04	Structural stability: issuer wording, 5-year validity, Government >50 yrs exception	`STRUCTURAL_STABILITY_TRIGGER`	☐
V-05	Affidavit-2: stamp amount, sworn-before list	`AFFIDAVIT_2_DSC_AND_STAMP`	☐
V-06	UG Computer Applications / Management FSR; PG Design / Applied Arts / HMCT FSR	`rules.json` Annexure-5	☐
V-07	Cadre ratios by stream; Planning/Architecture special rules	C-03	☐
V-08	Low-enrollment relaxation scope (which programmes)	C-05	☐
V-09	Faculty-deficiency (10%) and area-deficiency (10%) thresholds and consequences	`structural_thresholds`	☐
V-10	Adjunct/PoP: base text, 2025-26 changes, 2026-27 changes, stream scope	C-01	☐

#	Item	Where in your spec	Status
V-11	Annexure-4: PCs per department, PC:student ratio table, printers, A1 plotter, bandwidth tiers	C-11	☐
V-12	Page numbers for every `source_ref`	C-06	☐
V-13	EVC 30-min walkthrough clause; website 1–2 min video clause	`EVC_30MIN_WALKTHROUGH_VIDEO`	☐
V-14	MoU disclosure clause (18.18) and ABC applicability (18.15.xvii)	Annex-18 rules	☐
V-15	Minimum classroom/lab area (66 sq.m)	`structural_thresholds`	☐
V-16	TER definition and payment-verification path	`Bharatkosh_TER` adapter	☐
V-17	Which categories route to EVC	`workflow_routing`	☐
V-18	Supernumerary seats (25%) and "nil deficiency on self-disclosure" text	U-01 sample overlay	☐
V-19	Rounding practice for faculty computation (ask or infer from past LoDs)	C-02	☐
V-20	Definition of `S` for multi-shift / multi-year programmes	C-02	☐
V-21	Diploma formula in 2024-27	C-04	☐
V-22	Document validity rules (e.g., fire NOC validity vs academic session)	`FIRE_AND_OCCUPANCY_ELIGIBILITY`	☐
V-23	UDIN format; SHCIL verification mechanics (for adapter realism)	adapters	☐

9.3 Licence decisions (confirm each on the project's own page)

Component	Licence (to confirm)	Decision
pyHanko	MIT	Use
pikepdf	MPL-2.0	Use
pdfplumber / pdfminer.six	MIT	Use
pypdfium2	Apache-2.0 / BSD-3	Use
pypdf	BSD-3	Use
OpenCV (`opencv-python-headless`)	Apache-2.0	Use
zxing-cpp	Apache-2.0	Use
imagehash	BSD-2	Use
rapidocr-onnxruntime	Apache-2.0	Use
Tesseract	Apache-2.0	Use
onnxruntime	MIT	Use
FastAPI / Pydantic	MIT	Use
argon2-cffi, cryptography	MIT / Apache-2.0 + BSD	Use
rapidfuzz, selectolax	MIT	Use
reportlab, qrcode	BSD	Use
PyMuPDF	**AGPL-3.0 or commercial**	**Avoid**
Ultralytics YOLOv8	**AGPL-3.0 or enterprise**	**Avoid**; use Apache-licensed weights via ONNX

9.4 Test vectors (put these in `tests/golden/`)

ID	Input	Expected
T-01	`older_than(1996-01-01, 2026-01-01, 30)`	`False` (exactly 30 years)

ID	Input	Expected
T-02	`older_than(1996-01-01, 2026-01-02, 30)`	`True`
T-03	`valid_for_years(2020-01-01, 2025-01-01, 5)`	Boundary policy decides; `near_boundary == True` → Manual Review
T-04	`ug_cadre(S=1080, fsr=20)`	total 54; professor cadre 6 (5 + Principal); associate 12; assistant 36
T-05	`ug_cadre(S=1080, fsr=20, relax=True)`	total 41; 5 / 9 / 27
T-06	`max_intake_supported((4,10,30), fsr=20)`	720
T-07	`max_intake_supported((6,10,30), fsr=20)`	900
T-08	PoP 11/60 under 2024-25 stack	flagged (18.3% > 15%)
T-09	PoP 11/60 under 2025-26 overlay	passes (18.3% ≤ 20%)
T-10	PoP 14/60 under both	flagged (23.3%)
T-11	Plan area total ≈ 10.76 × declared total	"possible sq.ft/sq.m mix-up"
T-12	Role forged in login body	403 / ignored
T-13	Clean packet with CamScanner producer, no EXIF	`Manual Review Required` at most; never `Mismatch Detected`
T-14	Single Medium forensic signal	`Manual Review Required`
T-15	Medium (structural) + Medium (pixel) at same value	`Mismatch Detected`
T-16	Planner closure: inventory = planner output	no eligibility findings
T-17	`reproduce.py` on bundle; flip 1 byte in `facts.json`	exit 0; then non-zero
T-18	Overlay self-approval	403

9.5 Glossary

APH Approval Process Handbook · **EoA** Extension of Approval · **LoA/LoD/LoR** Letter of Approval/Deficiency/Rejection · **EVC** Expert Visit Committee · **SAC/SHC** Standing Appellate / Hearing Committee · **FSR** Faculty:Student Ratio · **PoP** Professor of Practice · **DSC** Digital Signature Certificate · **CCA** Controller of Certifying Authorities · **FAR** False-Accusation Rate (this document) · **Wilson interval** binomial confidence interval used for small samples · **Overlay** dated patch to the rule stack.

9.6 Changelog v2.0.0 → v2.1.0

Area	Change
Rules	Single frozen pack → base + addenda overlays, cycle-resolved, provenance, verified flags (C-06, C-10, U-01)
Regulatory content	Adjunct/PoP, cadre formula, relaxation, unverified FSR rows, Annexure refs (C-01…C-12)
Model	Completeness ⊥ verification status; roll-up semantics; claim evidence levels (C-09, C-31, U-05)
Forensics	Corroboration rule; calibrated severity; ELA supporting only; revision diff (C-22…C-25, P-02)
Verification	Tiers T0–T5, feasibility matrix, DigiLocker consent-pull design, SIMULATED watermark (U-06, P-01)
Security/privacy	Server-side roles, PAN digest, envelope encryption, authenticated page images, retention, chain checkpoints, sandbox (C-13…C-18)
Licensing/build	AGPL replacements, requirements, Docker (C-19, C-20)
Evaluation	Eight metrics, template shift, hard negatives, Wilson CIs, FAR, red-team matrix, time-and-motion (C-28, §6, U-12)
New features	Impact simulator, remediation engine, Intake Planner, Disclosure Auditor, reproducibility bundle, floor-plan reconciliation, consistency index, governed calibration (U-02…U-11)
Workflow	State machine enforcing human review (C-29)
Scope	Trimmed overlap with PS 26262; cut list (C-32, §8.5)

End of document.